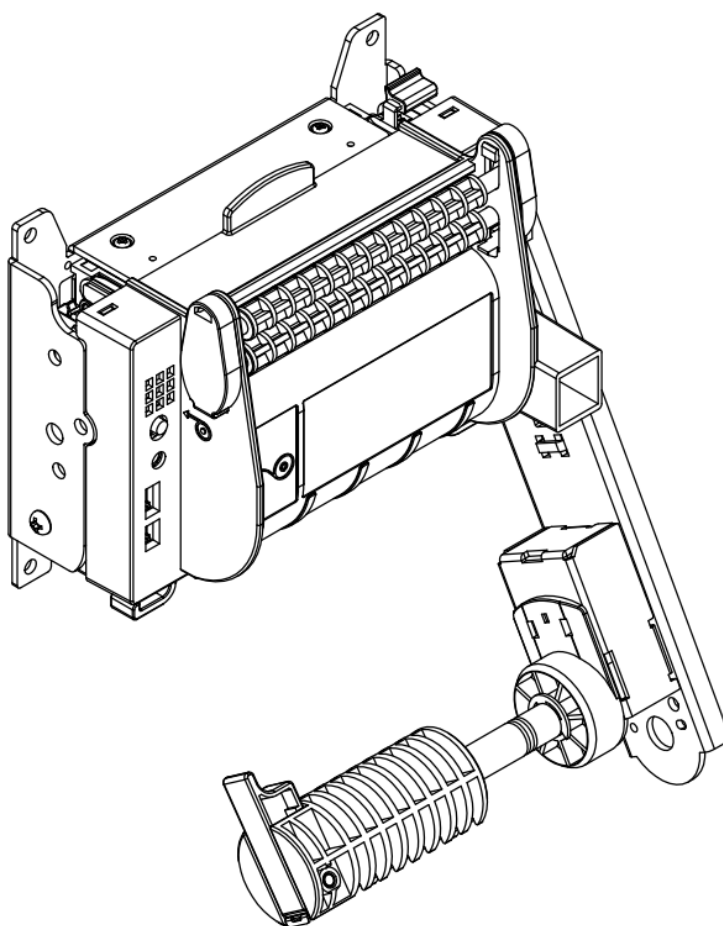


Product Specifications

NP-F309*

3 inch Face Mount Printer



Revision History

Ver.	Descriptions			Approval	Person in charge
	Page	Item	Revised points		
1		Newly released		Suzuki 2013.06.27	Abe 2013.06.21
2	-	Precautions	Adding and changing descriptions.	Hirano 2014.08.07	Abe 2014.07.31
		1D barcodes	Adding “CODE93” and “GS1 DataBar”.		
		How to set roll paper	Adding descriptions regarding "TEST PRINT".		
3		Precautions regarding printing operation	Adding descriptions	Osano 2016.12.26	Abe 2016.12.14
		Precautions regarding others	Adding “Firmware and hardware upgrades will be charged”.		
	2	Application	Changing descriptions in [Notes].		
	2	Model description	Changing descriptions.		
	4	Composition of deliverables	Changing descriptions.		
	5	Options	Adding descriptions.		
			Changing PS8 to PS11.		
	6, 7	Basic specifications	Changing descriptions.		
	8	Paper specifications	Changing descriptions in [Notes].		
	11	Print area 2)	Changing descriptions.		
			Adding [Notes].		
	15	Consumption current	Adding descriptions in [Notes].		
	22	Interface [USB (V2.0 FULL SPEED)]	Changing descriptions of *1.		
			Adding descriptions regarding NPI Integration Driver.		
	24	Serial connection wiring and flow control	Adding		
	34	ALARM lamp display and switching function	Changing descriptions.		
	Command Reference				
	4	Printer driver	Changing descriptions.		
			Adding descriptions regarding NPI Integration Driver.		
	27	GS d Dn	Adding descriptions.		
4	31	4.2 Processing errors	Deleting errors regarding power-supply voltage.		Watanabe 2022.01.12
	34	4.6 ALARM lamp display and switching function	Deleting errors regarding power-supply voltage.		

[Notes]

- The page numbers above are the page numbers in the revised version.

[VCCI Class A]

This device is a Class A information technology device. If this equipment is used in a home environment, radio disturbance may arise.

In this case, the user may be required to respond appropriately.

VCCI-A

[FCC Class A]

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Precautions

[General precautions]

Reprinting, copying, reproducing, and tampering all or a part of this document without permission is strictly prohibited.

- Descriptions in this document may be changed without prior notice. Please contact us for the latest information.
- Although we made assurance doubly sure in writing this product specifications, please contact us if you find any suspicious points, errors or what you noticed.
- Please check whether there are no problems for using this product by evaluating and testing it from the safety, operability, connectivity, and compatibility perspective before using it.
- Please note that we are not responsible for any failures or damage caused by the handling that is different from the descriptions in this document.
- Please note that we are not responsible for any damage caused by repairing or modifying the product by anyone other than NIPPON PRIMEX INC. and NIPPON PRINTER ENGINEERING INC.
- Please note that we are not responsible for any damage caused by using options or consumables that are not specified in this document.
- Please tell us when you find any potential malfunctions during the verification and evaluation.

[Precautions for using this product safe]

Incorrect handling will significantly reduce the performance of the product and may cause damage or injury. Please observe following precautions. Since there are detailed precautions described in other sections, please read this specifications document carefully before using the product

- If this product is used while it is in an abnormal state such as emitting smoke, a strange smell or sound, there is a risk of fire or electric shock. Please turn OFF the power switch and unplug the power supply cable from the outlet immediately.
- Please DO NOT modify this printer product. There will be a risk of injury, fire or electric shock.
- In order to avoid the risk of electric shock, please DO NOT install this product or connect any cables during a thunderstorm.
- Since there is a risk of electric shock, please DO NOT plug and unplug the power supply plug with wet hands.
- Since there is a risk of fire or electric shock, please DO NOT enter foreign substances into inside of this product or pour them on it.
- When water or other liquid comes inside this product, please turn OFF the power switch and unplug the power supply cable from the outlet immediately. If it is used continually, there will be a risk of fire or electric shock,
- Please be careful of power supply cable handling. Improper handling may cause injury, fire or electric shock,
 - Please DO NOT modify the power supply cable.
 - Please DO NOT put something heavy on it.
 - Please DO NOT bend, twist and tug at the cable.
 - Please DO NOT wire the power supply cable near the heating appliances.
 - Please DO NOT plug the power supply cable plug with adhering dust or dirt.
 - Please plug the power supply cable plug fully until it stops.

[Precautions regarding handling]

- Please DO NOT drop this product and give a strong shock to it.
- Please DO NOT put anything on this product. Heavy weight may damage this product.
- Please DO NOT use this product in following places since it will cause failures.
 - Where there is lots of dust, dirt or salt. Wet or oily places.
 - Slant places. Places with strong vibrations.
 - Places exposed to direct sunlight. Places near heating appliances and with 55 °C or higher temperature.
 - Places with -30 °C or lower or relative humidity 90% or higher.
 - Places where condensation is expected to occur due to sudden temperature changes.
 - Places where electromagnetic noise or corrosive gases are expected to be generated.
- Since the heating element of the thermal head and driver IC part are easily damaged, please DO NOT have static electricity, metal materials sand paper, etc. contact them.
- Static discharge prevention and body grounding must be made for dealing it in order to prevent any damage to electric heating element of the head or the IC, etc.
Please be sure to complete installing this product after connecting the frame ground of this product to the frame ground of the body of mating side for ESD measures, etc.
- Please DO NOT print when there is moisture on the head surface due to condensation, etc. Please be sure to dry the head completely before printing.
- Please DO NOT block the paper ejection slit during printing.
- Please DO NOT use volatile liquids such as thinner or benzine when cleaning.
- Please DO NOT pull paper from the paper ejection slit. It may cause failures.
- Using loose thermal roll paper may cause paper jams. Please remove any looseness before using.
- Please DO NOT apply excessive force to input terminals
- Please DO NOT have foreign substances mixed.

[Precautions regarding installation]

- Please refer to “External dimensions” in this document for installation. When installing this product, please put it horizontally not at slant.
- This product is equipped with an infrared reflection sensor. If the lights including infrared rays such as sunlight hit the sensor, malfunction may occur. This product must be installed so that it is NOT affected by disturbance light.
- This product is designed for indoor use. Please use it after fully evaluating and verifying its operation if it is used otherwise.
- Please DO NOT install this product in a place where it will be splashed with liquids or it will be mixed with foreign substances.
- When cooling this product with a fan, please make sure that the printer’s paper ejection slit does not serve an air intake or an exhaust port for the air. It will be a cause of premature failure by dust, etc.
- This product is not protected against dust or dirt. When used in a dusty place, the thermal head may be damaged or paper feeding may not run properly.
- This product should NOT be installed in a place where static electricity is easily generated and where it is exposed to vibration, electromagnetic fields, corrosive gas, sea breeze, rain, fog, or direct sunlight.

[Precautions to start using]

- The thermal head and the near area may get highly heated during or immediately after printing. Please DO NOT touch it so as not to get burnt. (Please be sure to start roll paper setting or thermal head cleaning after turning the power OFF and checking that the head part has become normal temperature.)
- Please NEVER open the thermal head cover during printing operations including cutting so as not to have the head and the cutter damaged.
- When taking the receipt after partial cut, please pull it to either right or left direction and tear it off.
- Please DO NOT pull out paper with the thermal head cover closed.
- Please DO NOT block the paper ejection slit and grab paper while doing printing operation.
- Please DO NOT pull paper during printing operation. It may cause paper jams or printing defects.
- Please avoid printing with no paper loaded. It can damage the thermal head and shorten its lifetime.
- In case of using roll paper other than recommended roll paper specified in this document, printing quality and lifetime of the thermal head may not reach the level guaranteed by us.
- Please DO NOT use label paper and copy paper.
- When this printer is not used, please turn the power OFF and unplug the power supply cable from the outlet.
- Please unplug the power supply cable or the interface cable with the connector part held instead of pulling the cable line part.
- Data may not be received due to paper-out.
- When this product is installed in a customer's device, paper sticking may occur after ejecting the printing result due to environmental conditions such as static electricity.
In that case, please consult us, as there may be cases in which static elimination measures can be considered.
- Since this product is not protected against water or dew drops, please DO NOT splash water on this product nor handle it with wet hands. They may be causes of damage, heat or fire due to the short circuit.
- Please turn OFF the power before connecting or removing connectors and NEVER disconnect it by pulling the cable without grabbing the connector body. Please DO NOT insert connectors diagonally, reversely or incorrectly. It will be a cause of failures.
- Please DO NOT disassemble NOR modify this product.
- Please DO NOT touch the cutter blade when the cutter is in operation or not.
- Please use the power supply conforming to the LPS (Limited Power Source) standard.
- Please make sure that the printer is connected to the power supply properly before supplying the power. When using an AC adapter, etc., please be sure to connect it to the printer before connecting the AC (primary) side. The same applies when supplying the power from the customer's system side.
- In order to prevent excessive current, please add a protection element and a suitable fuse (please refer to the "Power supply specifications" for the capacity.) to 24V power supply line supplying the power to this product.
- Please be sure to turn the power OFF before starting cleaning, removing jammed paper or other maintenance works.
- Please use AC adapter PS-11 (conforming to LPS standard) assigned by us. Please DO NOT use the power supply, the AC cable or the battery and the charger other than ones assigned by us.
- Please turn OFF the power, detach the battery and unplug the power supply plug when it is not used for a long time. Please remain the paper roll set in order to protect the platen roller.

[Precautions regarding quality]

- Operation is not guaranteed if an undefined control code or a command is sent to this product.
- With printing being started from the waiting status or intermittent printing in which printing and paper feeding get temporarily interrupted due to data queuing, etc., printing and paper feeding may jumble on 1 to 4 dots lines from the start. Please be careful especially when printing graphics, etc.
- Since If printing at a high printing rate is continued, some areas on printing surface may become black not by the printing data but by the stored heat in the head. Please evaluate and verify it sufficiently on the customer's side.
- When printing with high printing rate is continued, non-printing points may become black since the thermal head becomes heated with accumulated heat.
- Please DO NOT touch the heating element part of the thermal head because the dirty head may degrade the printing quality.
- For a dot or two dots immediately after cutting operation, printing may jumble and printing quality may deteriorate.
- Please DO NOT turn the power OFF unless an inevitable abnormal emergency case occurs. Data may be lost,
- Continuous paper feeding motor running for a long time generates heat and may affect the printer performance. To avoid the case, it is necessary to limit the continuous motor running time to 6 minutes for one cut with the interval time as long as the continuous motor running time period.
- If the cutter motor is driven continuously for a long time, the motor will generate heat and the required performance may not be achieved. Please be sure to observe the allowable frequency of the cutter. Even when the cutter is used within the allowable cutting frequency, please make the period with continuous cutting operations one minute at maximum and please take the interval time for the same duration as the period.
- Please set paper straightened with no slack.
- When selecting RTS / CTS in serial flow control, please be sure to connect RTS / CTS signal to the flow control signal on the host side, otherwise flow control will not function and may cause garbled characters or printing disarray.
- Since printing status change may cause unstable printing operation when printing the raster bit image while using the command [Print start / finish setting] and when obtaining the printer status, please obtain printer status not by the command [Printer status transmission] instead of the command [Automatic printer status transmission].
- After this printer has not been used for a long time, platen may be deformed and it may make printing quality deteriorate. In order to avoid this trouble, please feed paper for a while for making the platen adjusted before starting printing.

[Precautions regarding roll paper to be used]

- Please use thermal paper recommended by us or an equivalent product. If you use any other paper than the recommended one, printing quality and thermal head lifetime cannot be guaranteed. Please note that if thermal roll paper contains Na⁺, K⁺, Cl⁻, it can significantly reduce lifetime of the thermal head.
- Please store thermal roll paper in the dry, cool and dark storage.
- When gluing thermal paper, use a water-based glue (starch-based glue, synthetic glue, etc.). (Adhesive tape may discolor the thermal paper.)
- Thermal paper is specially processed paper whose surface has been specially treated with chemicals and it can develop black color through a thermochemical reaction. Please pay close attention to the following points.
 - Please DO NOT have printed thermal paper exposed to bright light for long periods of time.
 - Please avoid high temperature, humidity, liquids and sunlight.
 - Please DO NOT rub thermal paper hard with hard objects.
 - Please DO NOT place it near organic solvents.
 - Please DO NOT make it contact with polyvinyl chloride film, erasers or adhesive tape for a long time.
 - Please DO NOT stack with diazo or wet copies immediately after copying.
 - Please DO NOT touch thermal paper with sweaty hands. Fingerprints may be left on the surface and it will make the print result difficult to see.

[Precautions regarding others]

- This product is designed to be used with general electronic devices (computers, PC or devices used in offices, etc.). This product is not designed and guaranteed to be used with devices that require extremely high quality and reliability and to use with devices that their failures may directly endanger human body and life (atomic power control device, aerospace and aircraft device, transportation device, traffic signal device, ignition control device, medical device and various safety devices: hereafter called “specific application”). Users shall take full responsibility for using this product with such specific applications.
- Data cannot be long-term stored, permanently stored and saved since it is basically evanescent. NIPPON PRIMEX INC. is not responsible for any damage of data deletion or lost profits due to breakdown, repair or inspection.
- Specifications and shapes, etc. may be changed due to parts sharing or improvement of parts, etc. Please note that firmware, etc. may be changed.
- The coverage of warranty is limited within the delivered product itself and NIPPON PRIMEX INC. is not responsible for anything induced by the defect of the product and does not pay for any compensation for them.
- Since there is no galvanization coating on the end surface while this product uses the galvanized metal sheet, rust may occur on it. However, this does not impair the function.
- In case of disposal, please follow the regulations or rules of the local authorities.
- Please DO NOT conduct operation that is not described in this document. It may cause accidents or failures.
- Upgrades and specifications change of firmware and hardware will be charged.

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Appended reference documents (Please refer to documents below for commands and code tables.)

● Command Reference [NP-F309*] (D-F10119)	
Following code tables are described in the Command Reference in addition to the commands. Domestic character code table, Overseas character code table, Code Page 858, International character code table, Code Page 1250, Code Page 1251, Code Page 1252, Code Page 1254	
● Kanji code	[JIS C 6226·1983] (D-F10068)

1. Overview

1.1 Application

This document of product specifications is applied to 3 inch compact face mount KIOSK printer “NP-F309*”.

[Notes]

- Please refer to the Product Specifications separately for the specifications of options.
(Please note that descriptions in this document of printer's specifications may differ from specifications described in the Product Specifications of options.)
- Please refer to printer driver related documents when using the printer driver.
- Please note that usable roll paper, character types, the power supply and the environment, etc. may differ from those for other models.
- Specifications described in this document are subject to change without prior notice.

1.2 Model description

Basic models of this printer are categorized as follows.

NP-E3092D-**
(1) (2) (3) (4) (5)

(1) Type (fixed)

F: face mount type

(2) Mechanism (fixed)

3 : 3 inch (80mm)

(3) Paper holder type

Standard

2 : Roll paper with ϕ 120mm outer diameter at maximum [ϕ 120_Arm]

Specifications which can be considered as an OEM base

4 : Roll paper with ϕ 80mm outer diameter at maximum [ϕ 80_Arm]

(4) Interface

D : Both the serial (RS-232C) interface and the USB (V2.0 FULL SPEED) interface are equipped and selectable.

(5) OEM, etc. (factory default setting)

Without notation : original

1.3 Features

- 1) Face mount type printer model which can be attached to the front panel of machines directly.
- 2) Whether receipts will be ejected with the printing surface upward or downward can be selected at the timing of installation.
- 3) A paper holder with a cantilever type shaft realizes easy paper loading.
- 4) Roll paper can be set from both the right side and the left side. (It can be selected at the timing of installation.)
- 5) The position of roll paper can be selected from the upper area, the middle area and the lower area. (It can be selected at the timing of installation.)
- 6) Maximum printing speed is 150mm / sec.
- 7) With an auto cutter.
- 8) With paper near end detection function.
- 9) High quality printing.
- 10) Both serial interface and USB interface are equipped and selectable.
- 11) Various types of 1D barcodes are supported.
- 12) 2D barcode of QR code model2 is supported.
- 13) Applicable to various applications.
- 14) Several printer drivers are available.*1
- 15) Firmware*2 can be rewritten since the flash memory is adopted.
- 16) Up to 3 patterns of fixed bit image (logo, etc.) can be registered.
- 17) Easy to replace roll paper thanks to the auto-loading function.
- 18) An optional bezel is available.*3

*1 : Linux (CUPS driver) verification and evaluation has been done only for Ubuntu.

Please use the latest version of NPI Integration Driver.

*2 : "F/W" hereinafter stands for firmware.

*3 : Please refer to "1.5 Options".

1.4 Composition of deliverables

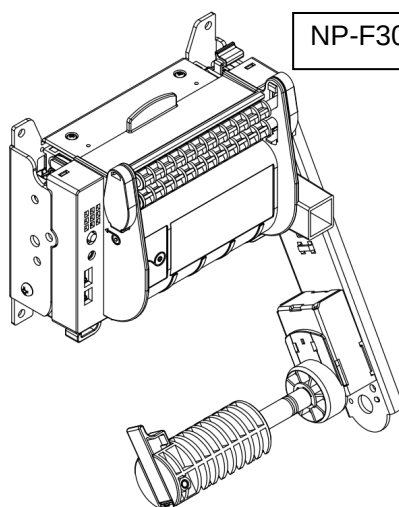
1) NP-F3092D (standard)

Components of this product are as follows.

Name	Descriptions	Quantity
NP-F3092D	3 inch Maximum outer diameter of roll paper is $\phi 120\text{mm}$. ($\phi 120_Arm$) Both the USB interface and the serial interface are equipped and selectable.	1

[Notes]

- There are no sample rolls of paper or other accessories.



NP-F3092D ($\phi 120_Arm$) [at the timing of shipping]

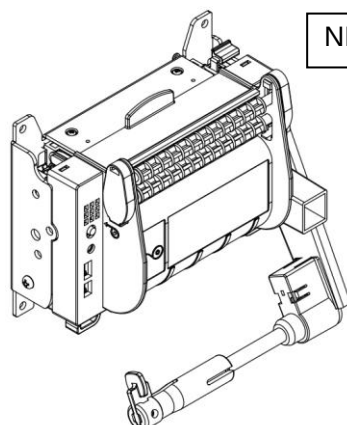
2) NP-F3094D

Components of this product are as follows.

Name	Descriptions	Quantity
NP-F3094D	3 inch Maximum outer diameter of roll paper is $\phi 80\text{mm}$. ($\phi 80_Arm$) Both the USB interface and the serial interface are equipped and selectable.	1

[Notes]

- There are no sample rolls of paper or other accessories.



NP-F3094D ($\phi 80_Arm$) [at the timing of shipping]

1.5 Options

The following options are available.

Since there are conditions and restrictions for using options, please read each product specifications and consider these options.

1) Peak time current responsive type AC adapter PS11-*2

This is an adapter for DC24V output.

Please refer to the product specifications D-F10183 and consider this option.

[Notes]

- Please evaluate and verify it with actual machines before printing with a high printing rate.

2) Bezel BEZ-430 provided specially for NP-F309*

A bezel for NP-F309* is available.

Please refer to the Product Specifications D-F10125 and consider this option.

2. Specifications

2.1 Basic specifications

Item			Specifications	
Printing method			Direct line thermal dot	
Number of dots / line			576 dots	
Dots density			8 dots / mm	
Maximum printing width			72mm	
Maximum printing speed *1			150mm / sec	
	Condition * Excluding communication time Head temperature 35°C or higher, print density 100% or less, optimized drive and printing rate 50% or less			
Maximum printing digits	Font A	12 x 24 dots	48 digits	
	Font B	9 x 17 dots	64 digits	
	Kanji	24 x 24 dots	24 digits	
Paper feeding pitch			0.125mm	
Character size	Font A	12 x 24 dots	1.50 x 3.00mm	
	Font B	9 x 17 dots	1.13 x 2.13mm	
	Kanji	24 x 24 dots	3.00 x 3.00mm	
Character types	Japanese		JIS C 6226・1983	double byte characters
			Domestic character code	single byte characters
			Overseas character code	single byte characters
			Code Page 858	single byte characters
			International characters	single byte characters
	Polish		Code Page 1250	single byte characters
	Russian		Code Page 1251	single byte characters
	Scandinavian		Code Page 1252	single byte characters
	Turkish		Code Page 1254	single byte characters
Character modification			Double width, double height, double width and double height, enhanced print, double strike print, inverted print, 90°clock-wise rotation, underlined	
Line feeding amount (default setting)			4.25mm (1/6 inch) *2	
Printing mode			Line mode Page mode	
Barcode	1D barcodes		UPC-A, UPC-E, JAN-13 (EAN-13), JAN-8 (EAN-8), CODE39, ITF, CODABAR, CODE128, CODE93, GS1 DataBar	
	2D barcodes		QR code model2	
Interface	USB		Conforming to V2.0 FULL SPEED	
	Serial		Conforming to RS-232C	
Cutting mode			Full cut, partial cut	
	* It is selectable by the command “[Full cut] 《ESC i》 ” and the command “[Partial cut] 《ESC m》 ”.			
Receiving buffer			Approx. 15K bytes	
Alarm display			ALARM LED	
Operation switch			FEED button RESET button	

*1 : Printing speed fluctuates depending on the condition.

*2 : It can be changed by the command "[Line feeding amount setting to the smallest paper feeding pitch unit] 《ESC 3 n》" described in the Command Reference.

Item		Specifications	
External dimensions * Excluding shipping form, stoppers, and roll paper	NP-F3092	Approx. 122.2mm (W) x 97.6mm (D) x 160.8mm (H)	
	NP-F3094	Approx. 121.2mm (W) x 73.0mm (D) x 123.5mm (H)	
Weight * Excluding roll paper	NP-F3092	Approx. 597g	
	NP-F3094	Approx. 520g	
Mounting orientation		Please refer to “5. Paper holder”.	
Individual carton	Dimensions		Approx. 224mm (W) x 169mm (D) x 168mm (H)
	Gross weight	NP-F3092	Approx. 830g
		NP-F3094	Approx. 750g
Master carton	Dimensions		Approx. 531mm (W) x 472mm (D) x 375mm (H)
	Gross weight (including 12 pieces)	NP-F3092	Approx. 12.9kg
		NP-F3094	Approx. 11.9kg

2.2 Paper specifications

1) Paper width and thickness

Paper width	80 ⁺⁰ ₋₁ mm
Paper thickness	From 59μm to 75μm

[Notes]

- Please use roll paper with the same paper core width as the paper width.

2) Form of roll paper

- Form has to be the roll one.

	Arm	Maximum outer diameter	Paper core inner diameter	Paper core outer diameter
NP-F3092	for φ120mm	φ120mm	φ25.4mm	φ33.4mm
NP-F3094	for φ80mm	φ80mm	φ12.0mm	φ18.0mm
			φ12.0mm	φ22.0mm

[Precautions regarding paper]

- Please use thermal roll paper.
- Please DO NOT use roll paper with the end of paper glued or taped.
- Please DO NOT use specially processed roll paper such as coreless or perforated.
- Please DO NOT use roll paper with the deformed paper core.
- Please DO NOT use roll paper which has the protruded paper core from the paper edge.
- Please DO NOT use roll paper which has been stored in high temperature and high humidity environment.
- Please use roll paper without loosening (slacks).
- Please use roll paper whose external side is used for the printing side [revolute].
(Involute roll paper is not available.)
- Please evaluate and verify it sufficiently before using pre-printed paper. Especially, please beware that there is a paper sensor behind the printing side of paper in the printer and there is a paper near end sensor on the side of Arm.
- Please DO NOT use roll paper which has been stored for a long time because printing quality will not be guaranteed.

3) Recommended thermal paper

Base paper model number	Paper thickness	Manufacturer	Recommended print density setting ^{*1}
TF50KS-E2D	59μm	NIPPON PAPER INDUSTRIES CO., LTD.	100% or higher
PD150R	75μm	Oji paper Co.,Ltd.	110% or higher
PD160R	75μm	Oji paper Co.,Ltd.	110% or higher

*1: The print density default setting is 100%.

The setting can be changed by the command “[Print density setting] 《GS ~ n》” described in the Command Reference.

(110% : n= [6E] h)

[Notes]

- When using paper other than recommended thermal paper, please evaluate and verify it sufficiently at the customers' own responsibility.
(Thermal paper containing less Na⁺ion, K⁺ion and Cl⁻ion suits well in order to prevent the galvanic corrosion.)
- Printing quality and darkness of the printing result may decline depending on the paper thickness, the paper surface coating or the environment of temperature and humidity. Please set the print density after evaluating and verifying the printing quality and darkness of the printing result with paper to be used and under the environment in which it will be used.
Print density setting can be done by using the command “[Print density setting] 《GS ~ n》” described in the Command Reference.
(Please set the print density at 130% at maximum.)
- When printing at a high printing rate under low temperature or high humidity environment, contamination on paper or condensation in this printer may occur due to vapors generated from paper. Please beware of water droplet not to drop on the thermal head. They may cause galvanic corrosion. When condensation occurs, please keep the power OFF until dew disappears.

4) Paper near end

How to set it

Position of the paper near end sensor is fixed and cannot be changed.

“Paper near end detection correction” can be set by “Memory switch setting”.

For details regarding “Paper near end detection correction” setting, please refer to “4.1 Function setting”.

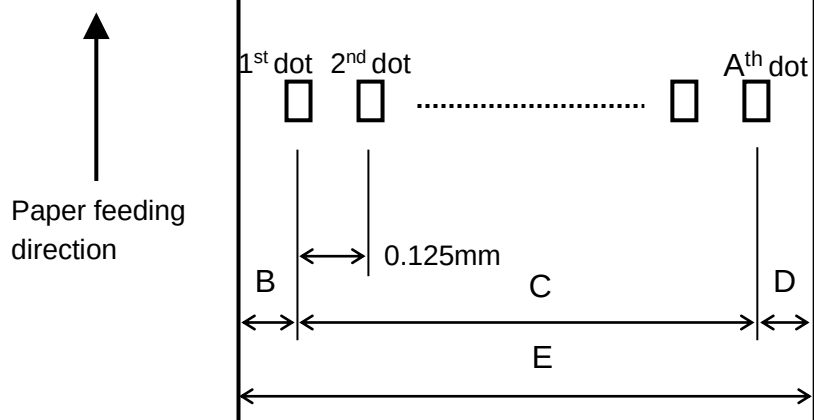
Please be sure to check the paper core outer diameter of roll paper to be used before setting it.

[Notes]

- Please consider these values for the paper near end detection outer diameter to be just reference since it varies depending on the paper type and paper thickness, etc.
- In case of selecting $\phi 22\text{mm}$ for the paper near end detection outer diameter with NP-F3094 (MS2-6: OFF) and the following (1) or (2) is caused, the sensor detects paper near end immediately.
 - (1) When small diameter roll paper having shorter diameter than approx. $\phi 28\text{mm}$ is newly loaded.
 - (2) When the power is turned OFF while remaining roll paper diameter is less than approx. $\phi 28\text{mm}$ and then the power is returned ON.

2.3 Print area

Printing side view



1) Code and descriptions

Code	Descriptions
A	The number of dots to be printed
B	Left margin
C	Print area
D	Right margin
E	Paper width

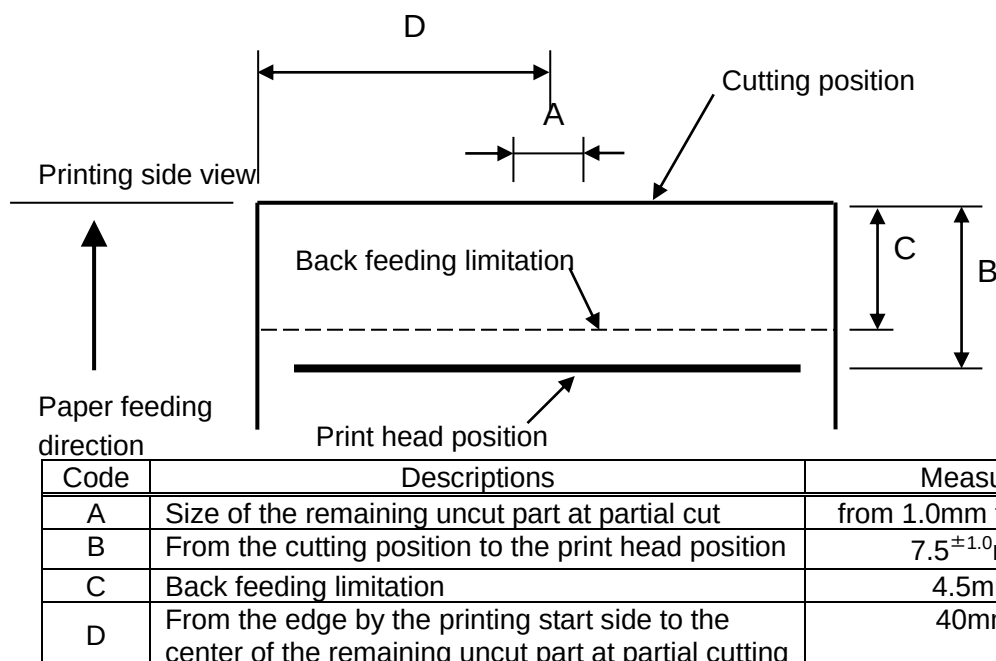
2) The relation between the paper width and the print area

A	B	C	D	E
576 dots	Approx. 4mm	$72^{\pm 0.2}\text{mm}$	Approx. 4mm	80^{+0}_{-1}mm

[Notes]

- There is a left margin of B and a right margin of D for a paper width of E in a print area of C.

2.4 Cutter specifications



- 1) Cutting system : Slide system
- 2) Cutting mode : Full cut / Partial cut

[Notes]

- It is selectable by using the command "[Full cut] 《ESC i》" and the command "[Partial cut] 《ESC m》".

3) Allowable cutting frequency : 20 cuts / minute (1 cut per 3 seconds)

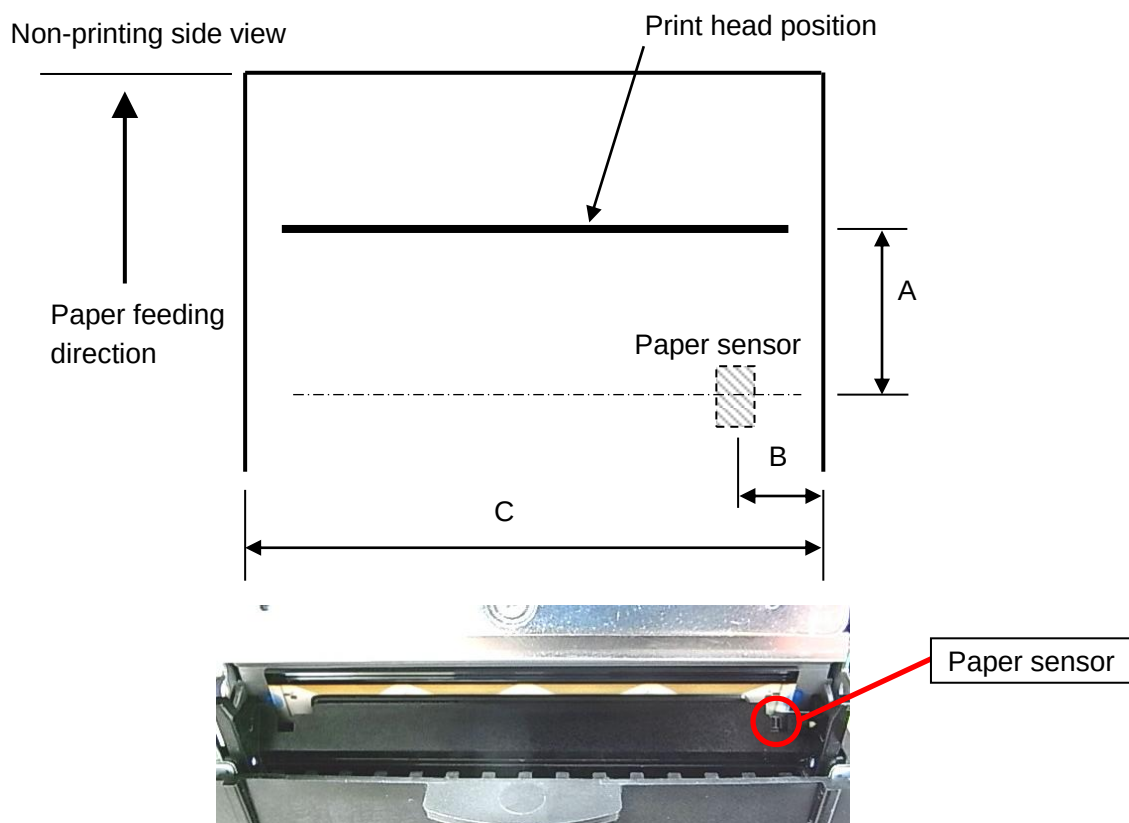
4) Paper thickness : Please refer to "2.2 Paper specifications".

5) Minimum cutting length : 30mm

[Notes]

- Please note that the head of the next line may be deformed after the platen gets heavily loaded by pulling paper strongly after partial cut, etc.
 - When printing after partial cut, please feed paper for 1mm (8 dots) or longer in order to prevent deformation of the upper part of the line.
 - Please tear off paper after partial cut by grabbing the tip of either left or right side of paper, pulling it horizontally toward the outside direction paper to the grabbed side.
 - Approx. 2mm paper feeding will be operated automatically after cutting in order to prevent paper jams. Therefore, measure of B in the above table will actually become 9.5 ± 1 mm.
 - Please DO NOT use it in a way which produces paper strips shorter than 30mm by cuttings. It will cause paper jams.
 - Partial cut is recommended for this product.
- Since cut paper may be stuck because of static electricity, etc., or may be caught in the paper ejecting slit, etc. depending on the environment when performing full cut, please use this product after verifying and evaluating it upon sufficiently.
- Since the multi-use cutter blade for partial cut and full cut is used and paper strips may appear by cuttings.

2.5 Paper sensor



[Notes]

- A sensor for paper detection is set in the body of the printer mechanism. (Please refer to the picture above.)
- Detecting whether paper is completely exhausted (paper end).
- Printing is stopped when paper end is detected.
- Please DO NOT use paper with the end of paper glued to the paper core since paper end cannot be detected.
- Please set new roll paper as soon as it detects paper end. _

1) Code and descriptions

Code	Descriptions
A	From the print head position to the sensor position
B	From the Sensor position to the left edge of paper
C	Paper width

2) Sensor position

A	B	C
$7.5 \pm 1 \text{mm}$	$4.0 \pm 0.5 \text{mm}$	80^{+0}_{-1}mm

[Notes]

- Please remove dust and paper powder adhering to the sensor periodically.
- Please DO NOT use roll paper which has pre-printed part in the range of 5mm left and right from the sensor position in order to prevent sensor's incorrect action.

2.6 Power supply specifications

1) Power supply input connector

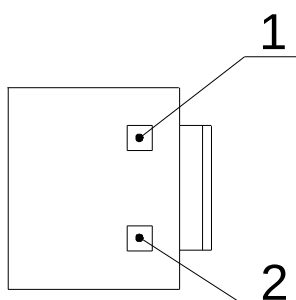
2-pin connector CN1

Please connect the adapter side connector described below to this connector (B2P-VH J.S.T.MFG.CO., LTD. or equivalent).

Adapter side connector : VHR-2N J.S.T.MFG.CO., LTD. or equivalent.

Pin No.	Function
1	DC+24V
2	GND

Connector figure (view from the mating surface)



2) Power-supply voltage : DC24V $\pm$ 5%

3) Consumption current

[Conditions] Partition drive : Optimization, Power-supply : DC24V, Temperature: 25°C, Print density: 100%

Printing rate	Approx. 18% (Please refer to the print example.) ANK rolling 50 lines (Repeat "A to Z", "0 to 9") + paper feeding 5 lines + full cut	100% (Please refer to the print example.) Printing length 20mm
Print contents	Font A 48 digits	Print width 72mm (576 dots)
Print example	$\longleftrightarrow$ 48 digits $\longrightarrow$ ABCD to 98ABCDEFGHIJKL MNOPQR to MNOPQRSTUVWXYZ YZ012345 to 0123456789 : : : YZ01 to WXYZ0123456789 ABCDEF to ABCDEFGHIJKL MNOPQRST to OPQRSTUVWXYZ	$\longleftrightarrow$ 72mm $\longrightarrow$ 
Reference consumption current	Average : approx. 1.0A Peak : approx. 4.5A	Average : approx. 3.2A Peak : approx. 4.5A

[Notes]

- Consumption current described above indicates the current during printing and it does not include the inflow current when the device is connected or electric power supplying is started.
- Settings of the print width and the partition drive have not been changed from factory default settings of this product.
- Printing rate ^{*1} = Number of energized printing dots / total number of dots on the head.
 ^{*1} : When printing with printing rate 60% or higher, please select optimization, dual partition drive printing or four partitions drive printing.
- Please DO NOT exceed the range of the power supply voltage under any conditions, as it may lead to deterioration or destruction immediately if the power supply voltage range becomes out of the range even for a moment.
- Please use the power supply with sufficient capacity in order to secure the printing quality since large peak current flows depending on the power voltage and printing contents. Furthermore, please be sure to connect all of the power supply relating wiring. Please use AWG #16 or others for the wiring material.
- The current capacity of thermal head itself is 10.2A. Since it does not include the current capacity of the connecting cable, please determine the number of dots being energized at one time in order not to exceed the current capacity.
- Operation may be unstable when power-supply line is long (2m or longer). Please shorten the power supply line. If it cannot be shortened, please connect cables near the printer and place an electrolytic capacitor (rated voltage 35V, electrostatic capacity 2200μF) between power supply and GND.
- Please wire the power line and the signal line so that they are not electromagnetically affected by the other power lines. Withstand voltage to use is 35V or more.
- Please be sure to set an element and a fuse to prevent excessive current flow.
- Please use the power supply which acquired LPS standards.
- Please DO NOT insert the connector diagonally, reversely or incorrectly. Furthermore, please DO NOT connect cables which may make nonstandard signals input as well as be sure to not make them input. It will be causes of the failure.
- Please DO NOT plug or unplug the active power supply cable.

2.7 Reliability specifications

Lifetime

(1) Thermal head

Anti-pulse resistance	: 100 million pulses
Anti-abrasion resistance	: 100km

(2) Cutter : 0.8 million cuts

(3) Lifetime definition

- Lifetime means the point which the printer is about to enter abrasion failure period.
- Conditions to satisfy lifetime are as follows.

Average printing rate	: 12.5%
Thermal paper	: Recommended paper
	Please refer to "2.2 Paper specifications".
Print density	: 100%
Environmental temperature and humidity	: 20°C, 60%

[Notes]

- Using paper other than recommended thermal paper may shorten the lifetime.

2.8 Environmental specifications

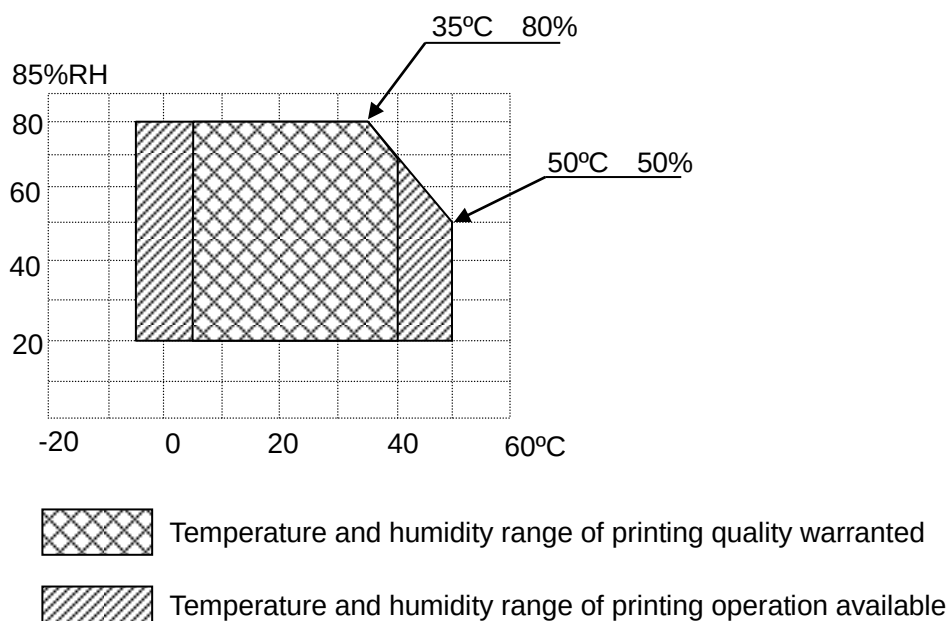
2.8.1 Operation environment

Temperature : -5°C to 50°C (The range of printing quality warranted is from 5°C to 40°C.)

Humidity : 20%RH to 80%RH

This range is under the condition without condensation and 80%RH has the premise that it is measured under 35°C.

The range of printing quality warranted and the range of printing operation available are described below.



2.8.2 Storage environment (excluding paper)

Temperature : -20°C to 70°C

Humidity : 10%RH to 90%RH

Under the condition without condensation

High temperature and high humidity : The upper limit is 40°C and 90%RH (without condensation).

2.8.3 Solder

Lead-free soldering is used for this product.

2.8.4 RoHS Directive

This product is conforming to RoHS Directive [2011/65/EU, (EU) 2015/863].

2.9 Standards

Standards for this product are as follows.

Standards	Class	Remarks
CE marking	Acquired	-
UL60950-1	Acquired ^{*1}	-
VCCI	Class A	Acquired
FCC	Class A	Acquired

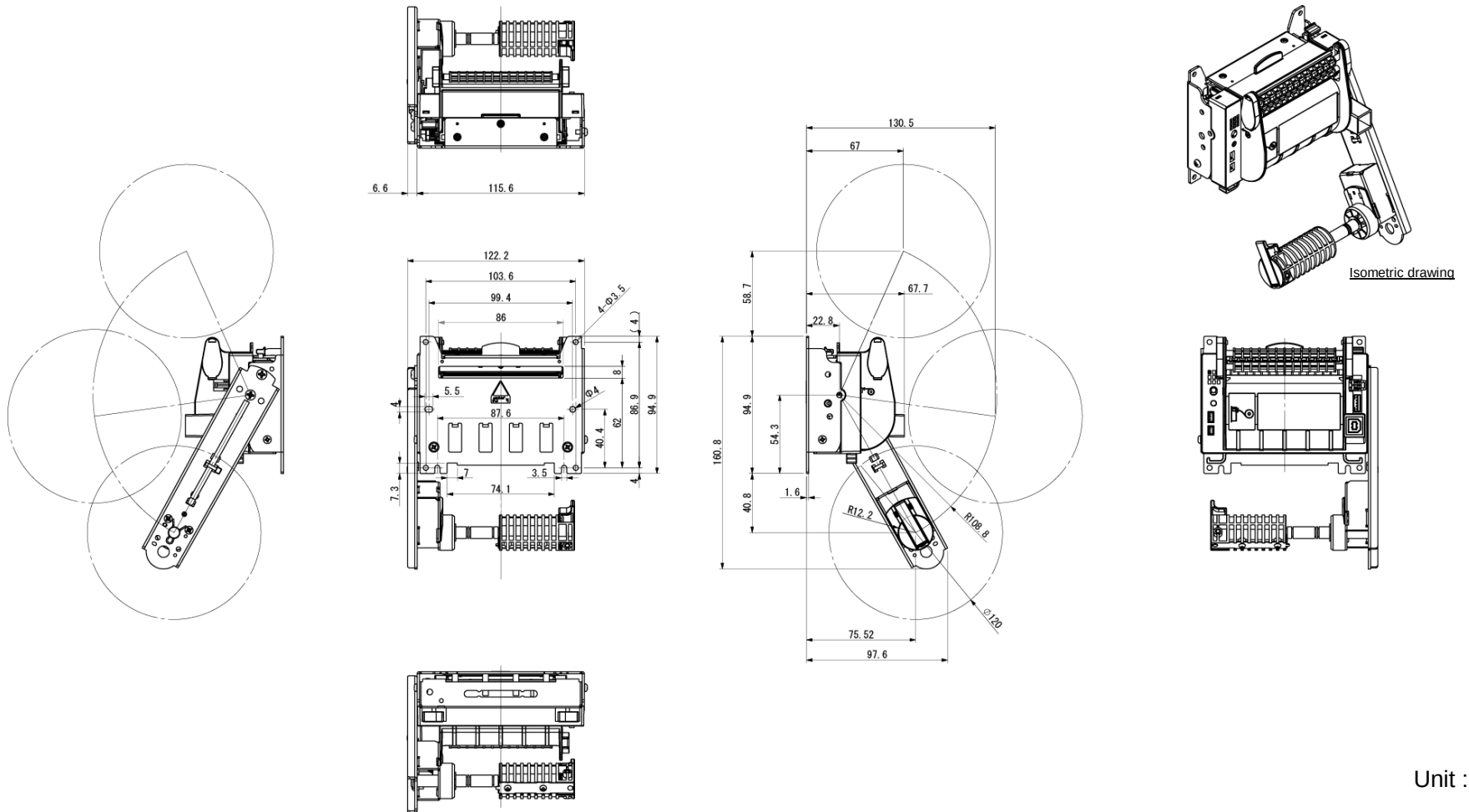
*1 : If the appearance at shipping is different from that of units delivered regularly, it will not conform to this standard.

[Notes]

- A standard model acquired standards above.

2.10 External dimensions

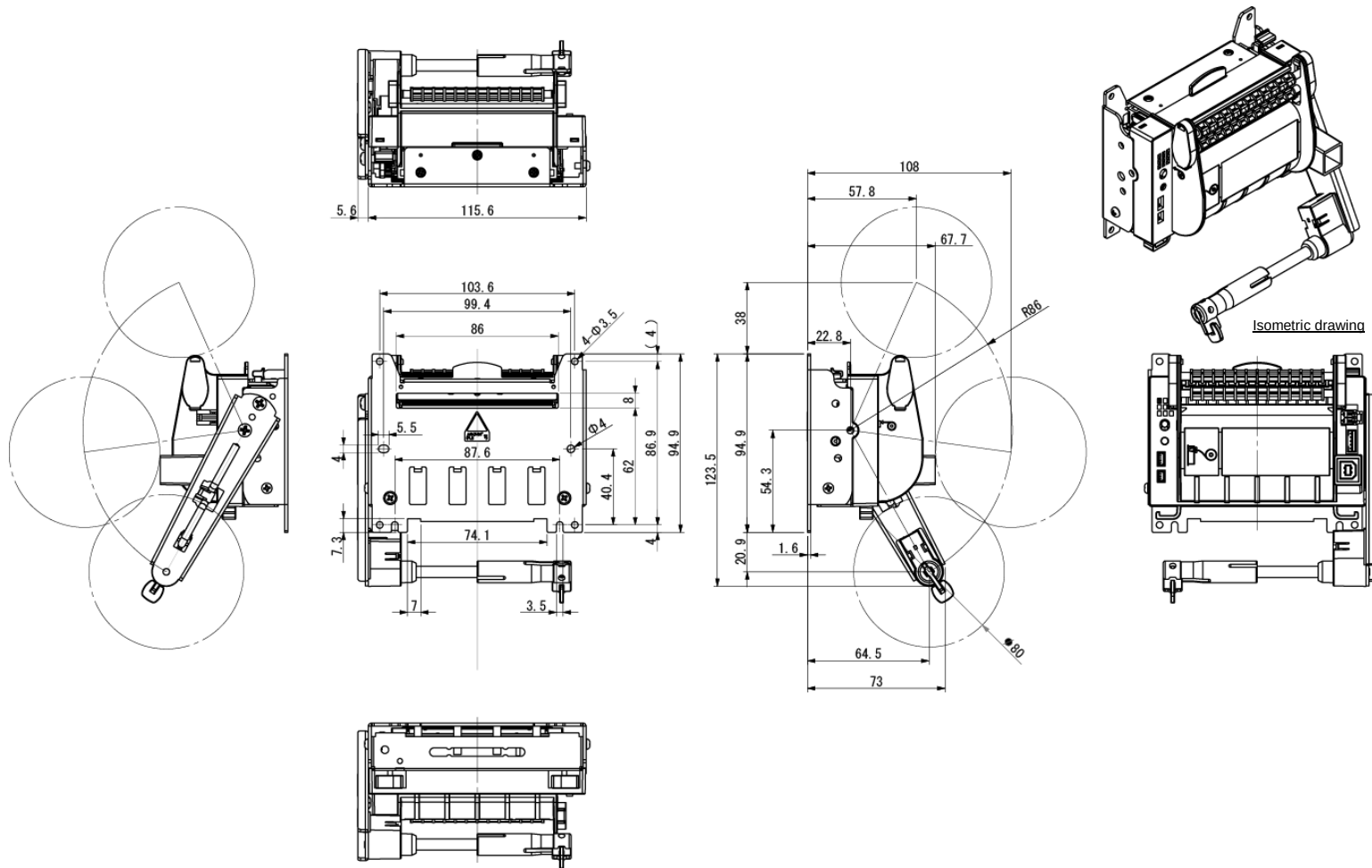
2.10.1 NP-F3092 (ϕ 120_Arm)



[Notes]

- By parts sharing or improvement, a shape may be changed, except outer dimensions, the positions of mounting holes or outlets etc. are specifically described.
- This picture shows the printer with the paper holder attached to the right side of the view from the paper insertion slit. (Roll paper will be loaded from the left side.)
- In case of attaching the paper holder to the opposite side, it will be symmetrical.

2.10.2 NP-F3094 (φ80_Arm)



Unit :mm

[Notes]

- By parts sharing or improvement, a shape may be changed, except outer dimensions, the positions of mounting holes or outlets etc. are specifically described.
- This picture shows the printer with the paper holder attached to the right side of the view from the paper insertion slit. (Roll paper will be loaded from the left side.)
- In case of attaching the paper holder to the opposite side, it will be symmetrical.

3. Interface

3.1 Interface [USB (V2.0 FULL SPEED)]

1) Version

V2.0 FULL SPEED (12Mbps)

2) Port

Upstream port (B jack)

3) Power supply

Self power

4) Reset function

This printer will be reset automatically when connecting the USB cable connected to the host.

5) Data receiving

USB data is received by a bulk-out transfer method. If the data is received even during printing and the remaining capacity of the receiving buffer becomes less than 1024 bytes, the NAK responses are continued until the processing proceeds and the remaining capacity of the receiving buffer reaches 1088 bytes or more. The maximum number of bytes that can be received in one packet is 64 bytes. Data is being received even when an error occurs.

6) Data transmitting

USB data is transmitted by a bulk-in transfer method. Response data is temporarily stored in the transmission buffer and a response with status data is sent when the bulk-in packet request from the host is received. If a bulk-in packet request is received when there is no data to be transmitted, only status data is returned. The maximum number of bytes that can be transmitted in one packet is 64 bytes.

[Notes]

- Please use one of drivers ^{*1} provided by us when using the USB interface.
- When performing the direct transmission, please monitor the remaining size of receiving buffer ^{*2} and DO NOT transmit the data whose size is beyond it.
- If using the direct communication without using our driver, please contact us.

*1 : Please use the latest version of NPI Integration Driver.

*2 : Receiving buffer remaining size auto-reply format (MS2-5 : OFF)

[FF] h+ [01] h+ [00] h+ [00] h+ [00] h+n ※ [00≤n≤0F] h

n= Receiving buffer remaining size (0 to 15K bytes)

3.2 Interface [serial (conforming to RS-232C)]

- 1) Synchronization mode : asynchronous mode
- 2) Data transmission speed : 2,400, 4,800, 9,600, 14,400, 19,200, 38,400, 57,600, 115,200bps (selectable by users)
- 3) Composition of one word
 - Start bit : 1 bit
 - Data bit : 7 bits or 8 bits (selectable by users)
 - Parity bit : an odd number, an even number or no parity (selectable by users)
 - Stop bit : more than 1 bit
- 4) Signal polarity (RS-232C)
 - Mark = logical"1" / OFF (-3V ~ -12V)
 - Space = logical"0" / ON (+3V ~ +12V)
- 5) Received data (RXD signal)
 - Mark = 1
 - Space = 0
- 6) Transmitted data (TXD signal) ^{*1}
 - Mark = 1
 - Space = 0
 - During XON/XOFF control
 - 《DC1》 [11] h code, XON : Possible to receive data ^{*2}
 - 《DC3》 [13] h code, XOFF : Impossible to receive data ^{*3}
- 7) Receiving control (RTS signal)
 - Mark : Impossible to receive data ^{*4}
 - Space : Possible to receive data ^{*2}
- 8) Transmitting permission (CTS signal)
 - Mark : Impossible to transmit data
 - Space : Possible to transmit data

^{*1} : Data transmission will not be operated during receiving data. In addition, data transmission will be operated after stopping receiving data when receiving data sequentially. Furthermore, the timing of data transmission may change depending on the conditions during printing or receiving data.

^{*2} : It will occur after turning the power ON, finishing self-diagnostic printing or completing software resetting and when releasing receiving buffer full or the firmware is rewritten.

^{*3} : It will occur when the receiving buffer is full, after receiving the command "[Memory switch setting and set contents printing] 《GS M n d1 d2》", after receiving the command "[Software reset] 《DC1》" or when the firmware is rewritten.

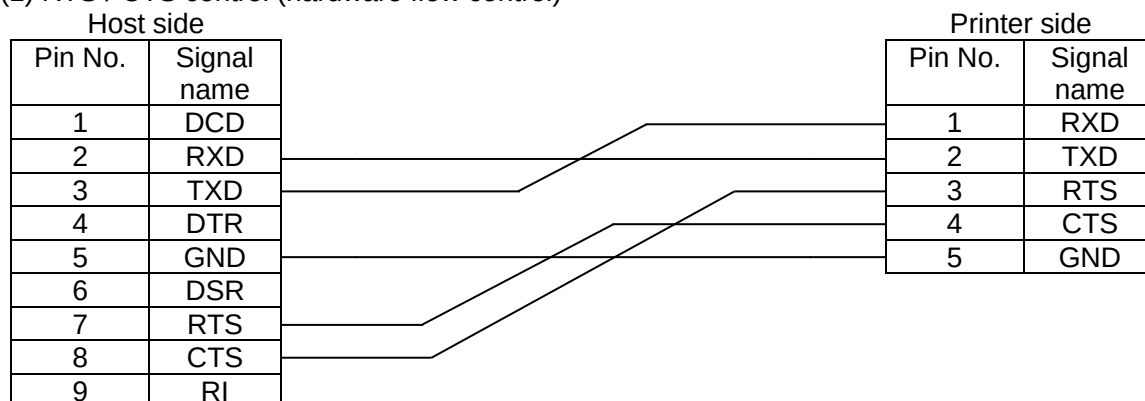
^{*4} : It will occur when turning the power OFF, self-diagnostic printing is executed, software resetting is performed or the receiving buffer is full, after receiving the command "[Memory switch setting and set contents printing] 《GS M n d1 d2》" or the firmware is rewritten.

[Serial connection wiring and flow control]

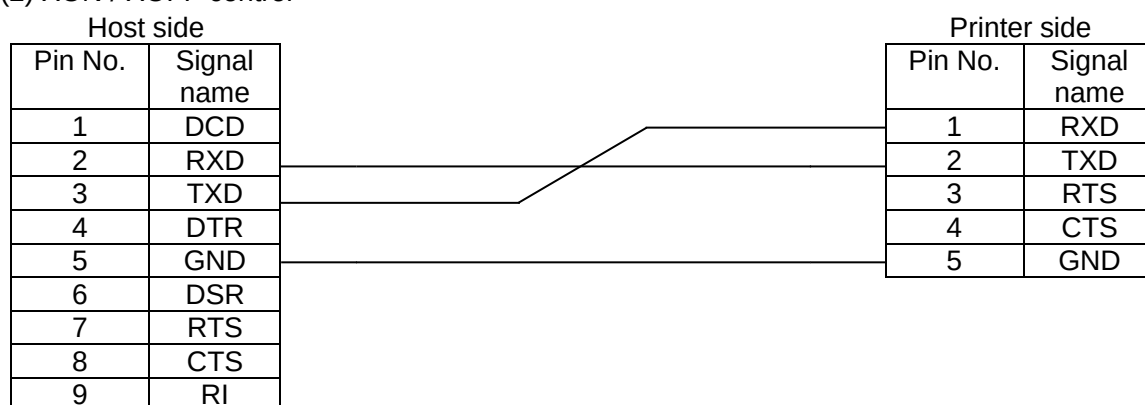
1) Connection wiring diagram

(AT compatible machines) JEC-9S (J.S.T.MFG.CO., LTD.) or equivalent

(1) RTS / CTS control (hardware flow control)



(2) XON / XOFF control



[Notes]

- The cable length must be within 3m.
- Please be sure to connect GND.
- Please be sure to shield the signal line. At that time, please DO NOT shield the power supply and signal lines together.
- Please check and verify this product sufficiently before using it actually.

2) Flow control

(1) RTS / CTS control (hardware flow control)

The RTS signal is marked and become unable to receive in the following cases.

- From power-on or resetting to the timing when the communication becomes possible after initialization processing.
- During self-diagnostic printing.
- When the receiving buffer is full.
- When defining / registering data to the printer.

(2) XON / XOFF control

XON / XOFF transmission timings are as follows.

XON transmission	• When the communication becomes possible after power-on or resetting. • When the receiving buffer full is released. • When data definition / registration to the printer are finished.
XOFF transmission	• When the receiving buffer is full. • When defining / registering the data to the printer.

3.3 Connector signal table

1) CN1 : Power input connector (2-pin connector)

Please refer to [2.6 Power supply specifications].

2) CN3: USB data signal input connector

Printer side : B jack CU02SCV1000 (CviLux Corporation) or equivalent

Mating side : B plug or equivalent

Pin No.	Signal name	Input / Output	Function	Remarks
1	VBUS	Input	Power line	Non-twisted power line
2	D-	Input / Output	Data line	Twisted pair signal line
3	D+	Input / Output	Data line	Twisted pair signal line
4	GND	—	Power line	Non-twisted power line
Shell	Shield	—	—	—

[Notes]

- Please use the USB cable which is conforming to the standard (FULL SPEED).
- Operations cannot be guaranteed if a nonstandard cable is used.

3) CN4 : Serial data signal input connector

Printer side : B5B-PH-K-S (J.S.T.MFG.CO., LTD.) or equivalent

Mating side : PHR-5 (J.S.T.MFG.CO., LTD.) or equivalent

Pin No.	Signal name	Input / Output	Function	Remarks
1	RXD	Input	Serial receiving data	—
2	TXD	Output	Serial transmitting data	—
3	RTS	Output	Receivable signal	—
4	CTS	Input	Transmitting permission signal	—
5	GND	—	GND for signal	—

[Notes]

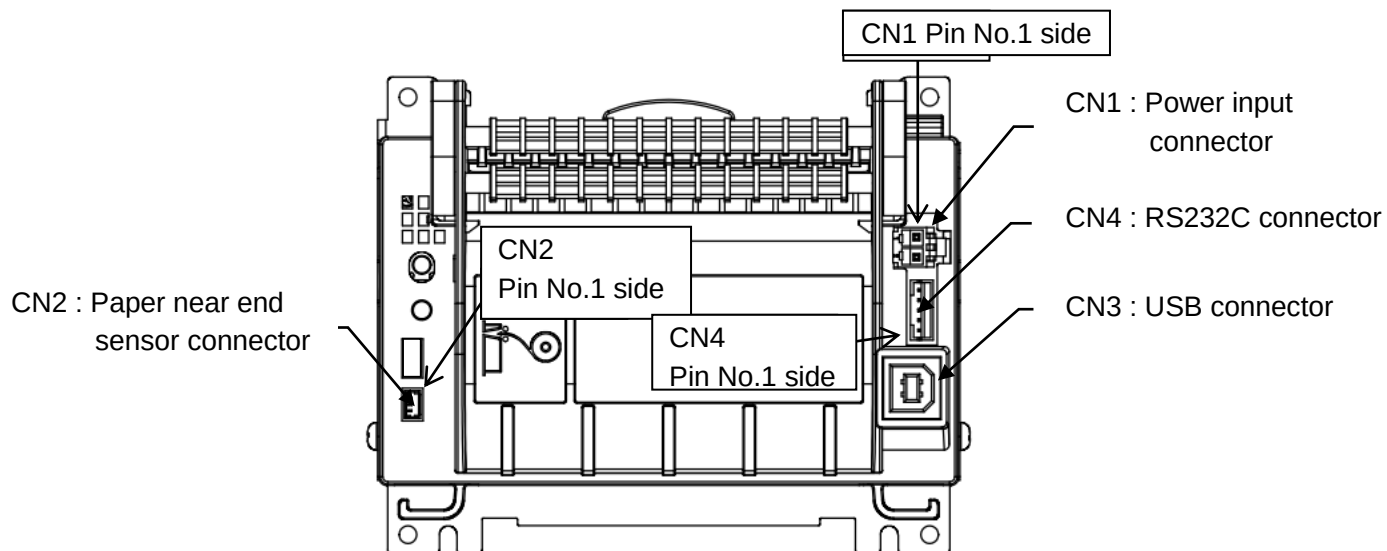
- Before using this product actually, please be sure to verify and evaluate it sufficiently on the user's side.
- Please make sure that the common current does not flow to the GND for signal.

4) CN2 : Paper near end sensor connector

“Standard paper near end sensor” is connected to this connector.

[Notes]

- Please DO NOT connect other than “standard paper near end sensor” to this connector.
- Please be sure to set MS2-4 to “Effective” before using it.



4. Function

4.1 Function setting

4.1.1 Switch setting

1) Memory switch MS1

	Function description	ON	OFF	Factory default setting	
				NP-F3092D (φ120_Arm)	NP-F3094D (φ80_Arm)
MS1-1	Communication setting	Please refer to the table 1.		OFF	OFF
MS1-2				OFF	OFF
MS1-3				OFF	OFF
MS1-4	Serial communication speed	Please refer to the table 2.		ON	ON
MS1-5				OFF	OFF
MS1-6				ON	ON
MS1-7	Serial flow control *1	XON / XOFF	RTS / CTS	OFF	OFF
MS1-8	Auto cutter control	ineffective	effective	OFF	OFF

*1: Flow control during XON / XOFF controlling

(1) During XON / XOFF control, all statuses will be transmitted as the data converted to ASCII into “ in 0 x FF 0 x FE 0 x 00”.

(2) ASCII conversion method is as follows.

Higher 4 bits = 0 x 30 + higher 4 bits >> 4 (bit shift)

Lower 4 bits = 0 x 30 + lower 4 bits

Therefore, they will be converted to the values in the table below.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	00	10	20	30	40	50	60	70	80	90	:0	:0	<0	=0	>0	?0
1	01	11	21	31	41	51	61	71	81	91	:1	:1	<1	=1	>1	?1
2	02	12	22	32	42	52	62	72	82	92	:2	:2	<2	=2	>2	?2
3	03	13	23	33	43	53	63	73	83	93	:3	:3	<3	=3	>3	?3
4	04	14	24	34	44	54	64	74	84	94	:4	:4	<4	=4	>4	?4
5	05	15	25	35	45	55	65	75	85	95	:5	:5	<5	=5	>5	?5
6	06	16	26	36	46	56	66	76	86	96	:6	:6	<6	=6	>6	?6
7	07	17	27	37	47	57	67	77	87	97	:7	:7	<7	=7	>7	?7
8	08	18	28	38	48	58	68	78	88	98	:8	:8	<8	=8	>8	?8
9	09	19	29	39	49	59	69	79	89	99	:9	:9	<9	=9	>9	?9
A	0:	1:	2:	3:	4:	5:	6:	7:	8:	9:	::	::	<:	=:	>:	?:
B	0;	1;	2;	3;	4;	5;	6;	7;	8;	9;	::	::	<;	=;	>;	?;
C	0<	1<	2<	3<	4<	5<	6<	7<	8<	9<	:<	:<	<<	=<	><	?<
D	0=	1=	2=	3=	4=	5=	6=	7=	8=	9=	:=	:=	<=	=	>=	?=
E	0>	1>	2>	3>	4>	5>	6>	7>	8>	9>	:>	:>	<>	=>	>>	?>
F	0?	1?	2?	3?	4?	5?	6?	7?	8?	9?	:?	:?	<?	=?	>?	??

* They are displayed as character strings in this table.

(3) Contents specified in “” apply to all data including the standard status and the status header “(0 x FF)”.

(4) Printer disregards the command “ESC s [FE] h”.

(5) The driver and the tool do not issue “ESC s [FE] h” command.

Table 1 Communication setting

Interface	Bit length	Parity setting	MS1-1	MS1-2	MS1-3	Factory default setting
Serial	8 bits	Nil	OFF	OFF	OFF	○
			ON	OFF	OFF	—
		An odd number	OFF	ON	OFF	—
		An even number	ON	ON	OFF	—
	7 bits	Nil	OFF	OFF	ON	—
		An odd number	ON	OFF	ON	—
		An even number	OFF	ON	ON	—
Reserve	-	-	ON	ON	ON	—

[Notes]

- Please DO NOT set all of MS1-1, MS1-2 and MS1-3 to ON at the same time.
- Regarding interface

Even during the serial communication setting, printer's interface will become USB automatically when the printer detects VBUS signal after USB cable is connected.

In this case, once printer's interface becomes USB, even if the USB cable is disconnected, since it will never be returned to serial interface automatically, please turn the power OFF and ON for returning the setting to serial interface.

Table 2 Serial communication speed

Serial communication speed	MS1-4	MS1-5	MS1-6	Factory default setting
115,200 bps	OFF	OFF	OFF	—
57,600 bps	ON	OFF	OFF	—
38,400 bps	OFF	ON	OFF	—
19,200 bps	ON	ON	OFF	—
14,400 bps	OFF	OFF	ON	—
9,600 bps	ON	OFF	ON	○
4,800 bps	OFF	ON	ON	—
2,400 bps	ON	ON	ON	—

2) Memory switch MS2

	Function description		ON	OFF	Factory default setting	
					NP-F3092D (φ120 Arm)	NP-F3094D (φ80 Arm)
MS2-1	Japanese Kanji code		Shift JIS	JIS	OFF	OFF
MS2-2	Reserve		-	-	OFF	OFF
MS2-3	Reserve		-	-	OFF	OFF
MS2-4	Paper near end * ¹		ineffective	effective	OFF	OFF
MS2-5	Receiving buffer remaining size auto-reply (USB) * ²		ineffective	effective	OFF	OFF
MS2-6	Paper near end detection correction	NP-F3092D * ³	φ39mm	-	ON	-
		NP-F3094D * ⁴	φ28mm	φ22mm	-	OFF
MS2-7	Reserve		-	-	OFF	OFF
MS2-8	Reserve		-	-	OFF	OFF

*1 : Please set MS2-4 to ON when the paper near end is not connected. Standard configuration does not include it.

*2 : Please beware that the printing trouble of character garbling, etc. may occur when a large amount of data is transmitted at once during "MS2-5 : ON" setting.

*3 : When using NP-F3092D (φ120_Arm), please be sure to set MS2-6 to ON.

*4 : When using NP-F3094D (φ80_Arm), please refer to "2.2 4) Paper near end".

In case of "MS2-6 : OFF" setting, the detecting signal will become ON and it will become the paper near end after detecting the paper near end is detected by the sensor and feeding paper for approx. 3.7m. However, when it is detected right after turning the power ON and roll paper change after detection, it will become the paper near end status immediately.

In case of "MS2-6 : ON" setting, the detecting signal will become ON and it will become the paper near end immediately after the paper near end is detected.

Please use the value of paper near end detection outer diameter as a guide to the standard since it varies depending on the paper type and paper thickness, etc.

[Notes]

- Please be sure to set MS2-2, MS2-3, MS2-7 and MS2-8 to OFF.

4.1.2 How to set memory switch by manual operation

When setting (changing) the memory switch by manual operation, please conduct following operations with ready to print status.

1) Steps to enter the setting mode

- (1) Setting roll paper while the printer unit is closed and the power is OFF.
- (2) Keeping FEED button pressed units (3) and (4) is completed.
- (3) Turning the power ON and checking whether the printer is activated.
- (4) Opening the printer unit and closing it.
- (5) Releasing the pressed FEED button.
- (6) Setting mode will start and comments for several lines including the message
 “** MEMORY SW SETTING MODE **” will be printed.

2) How to set each switch

Please set each switch from MS1-1 to MS2-8 to “ON” or “OFF” sequentially with printed result checking in the setting mode.

Setting it to “ON”	Please press the FEED button and keep it pressed for longer than 1 second.
Setting it to “OFF”	Please press the FEED button and release it within 1 second.

Memory switch setting will be completed by repeating the operation above 16 times.
 Software reset will be performed and the latest setting will become effective immediately after the set result list is printed.

[Notes]

- The setting operation can be stopped in the middle of setting by opening the printer unit, pressing the FEED button once and closing the printer unit.
 By doing this, setting to the middle will be set and all MS switches after that will be set to OFF.

4.1.3 How to set memory switch by online command operation

The setting will be performed and changed by the following command from the host after paper is set and the printer will become ready to start online printing properly. When the printer receives the setting command normally, software reset will be performed and the latest setting will become effective immediately after the set result list is printed.

1) The command for setting

The command “[Memory switch setting and set contents printing] 《GS M n d1 d2》 ”
 The memory switch MS1 and MS2 can be set and set contents for memory switch MS1 and MS2 can be printed by using this command.

[Notes]

- Please refer to “4.1.1 Switch setting” for details and precautions regarding each memory switch.
- Please refer to the command “[Memory switch setting and set contents printing] 《GS M n d1 d2》 ” in the Command Reference for memory switch setting by the command.
- The command “[Memory switch setting and set contents printing] 《GS M n d1 d2》 ” can also be transmitted by using “Nii Printer Tool”, etc.

4.1.4 Self-diagnostic printing

- 1) Items which can be checked by self-diagnostic printing completion and the printed result
 - FW version
 - USB serial number
 - Results of memory switch setting

- 2) Starting and finishing self-diagnostic printing

Self-diagnostic printing will be started by turning ON the power with the FEED button pressed and releasing the FEED button after printer mechanism initialization.

It will be finished after printing the preset printing patterns.

The printer is offline while the self-diagnostic printing is being printed.

[Notes]

- USB serial number is printed when it has been set by the command “[USB serial No. setting] 《GS U Dn》 ” described in the Command Reference.

4.2 Processing errors

1) Details of error detection

ALARM LED ^{*1} status	Name	Status	Status information	How to release it
-	Communication error	RS-232C communication error Parity Overrun Framing	-	Adjusting communication condition.
Off	Normal	Normal	bit0-7 "0"	
Off	Printing start status	Printing start setting by the command "[Print start / finish setting] 《GS G n》" (This is not an error.)	bit7 "1"	Printing finish setting by the command "[Print start / finish setting] 《GS G n》"
-	Unused	-	bit6 "0"	-
-	Unused	-	bit5 "0" (always 0)	-
Blink	Auto cutter error	Paper jam around the cutter	bit4 "1"	Opening the printer unit, removing jammed paper and closing the printer unit.
Blink	Abnormal temperature of head	When the head temperature rises to above approx. 70°C	bit3 "1"	Automatically reset when the head temperature falls to below approx. 60°C.
Light	Paper end	Paper-out	bit2 "1"	Setting new roll paper.
Light	Printer unit open	Printer unit open	bit1 "1"	Closing the printer unit.
Blink	Paper near end	Paper near end is detected by the sensor during remaining paper amount detection. (MS2-4 : When paper near end detection is effective.)	bit0 "1"	Setting new roll paper.

*1: Please refer to [4.6 ALARM lamp display and switching function] for ALARM LED display patterns.

[Notes]

- The printer stops all ongoing operations when one of errors above except "Communication error", "Paper near end" or "Printing start status" is detected. (Printing operation or paper feeding by the FEED button cannot be performed as well.)
- Auto-loading will not be performed when errors above except "Paper near end", "Paper end" and "Printing start status" are detected.
- An error bit in the status information is "1".
- When "Paper end" is released by the auto loading, paper will be fed and full cut will be performed.

4.3 Buffer full printing

When the printing data for one line ^{*1} is received, the printer will operate line feeding automatically based on the amount specified by the command “[Line feeding amount setting to the smallest paper feeding pitch unit] 《ESC 3 n》” described in the Command Reference.

*1 : The amount of buffer full data (maximum number of digits for printing) differs depending on the size of each ANK・Kanji (Please refer to “2.1 Basic specifications”).)

4.4 Drive selection

Fixed partition (no-partition, dual partitions or four partitions) or optimization can be selected by the command “[Partition drive selection] 《GS % n》” described in the Command Reference. Please select it after taking the consumption current by printing rate into consideration. (Please refer to “3) Consumption current” in “2.6 Power supply specifications”).

1) Partition drive selection

Please refer to the command “[Partition drive selection] 《GS % n》” described in the Command Reference.

	No-partition	Dual partition	Four partitions
The number of dots being energized at one time	576 dots	288 dots	144 dots

[Notes]

- When other than no-partition is selected, printing speed will be declined. Furthermore, one dot height space line at maximum may appear depending on the printing contents (with printing rate changing for example).

2) Optimization

It switches the partition drive selection to “high speed”, “standard”, “current saving (low)” or “current saving (high)” according to the total printing number of dots per line.

	High speed	Standard	Current saving (low)	Current saving (high)
The number of dots being energized at one time	288 dots	144 dots	72 dots	48 dots

[Notes]

- When selecting optimization, printing speed changes depending on the printing rate and the sound during the printing may change.
- When selecting optimization, one dot height space line at maximum may appear. Furthermore, when selecting optimization, the printing quality may deteriorate or the sound during the printing may change.

4.5 How to select double-byte characters and single-byte characters

Language font	How to switch
Japanese (JIS C 6226·1983)	The command “[Kanji mode specifying] 《FS & 》”, the command “[Kanji mode releasing] 《FS .》” or Shift JIS code switching
Polish (Code Page 1250)	No switching (only single-byte characters)
Russian (Code Page 1251)	No switching (only single-byte characters)
Scandinavian (Code Page 1252)	No switching (only single-byte characters)
Turkish (Code Page 1254)	No switching (only single-byte characters)

4.6 ALARM lamp display and switching function

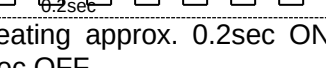
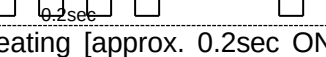
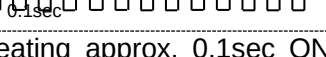
Operation panel part including following devices is provided on the printer's body.

1) ALARM LED (red)

Statuses of this printer are displayed according with display patterns of ALARM lamp.

It blinks, lights and then goes out when writing to the flash ROM is performed.

Display patterns of ALARM lamp are described in the following table.

Display patterns	Printer status	Priority (7: high ~ 1: low)
ON OFF 	Normal Printing (receiving) can be done.	1
Off ON  OFF  Repeating approx. 0.2sec ON and approx. 0.2sec OFF	Paper near end	2
ON  OFF 	Paper-out	3
On 	The printer unit is opened.	4
ON  OFF  Repeating approx. 2.2sec ON and approx. 2.2sec OFF	Head temperature abnormal (above approx. 70°C) or abnormal head connection	5
ON  OFF  Repeating [approx. 0.2sec ON and approx. 0.2sec OFF] ×3, approx. 1 sec ON and approx. 0.2 sec OFF	Auto cutter error	6
ON  OFF  Repeating approx. 0.1sec ON and approx. 0.1sec OFF	F/W writing mode	7

2) FEED button [for paper feeding]

The button for feeding paper to the forward direction.

This button is also used for self-diagnostic printing and memory switch setting.

[Notes]

- "Effective" or "ineffective" can be selected by the command "[FEED button effective / ineffective] 《ESC c 5 n》".
- When the FEED button is pressed and then released, full cut will be performed after approx. 60mm paper feeding. (Paper is being fed while FEED button is kept pressed and full cut will be performed when it is released.)

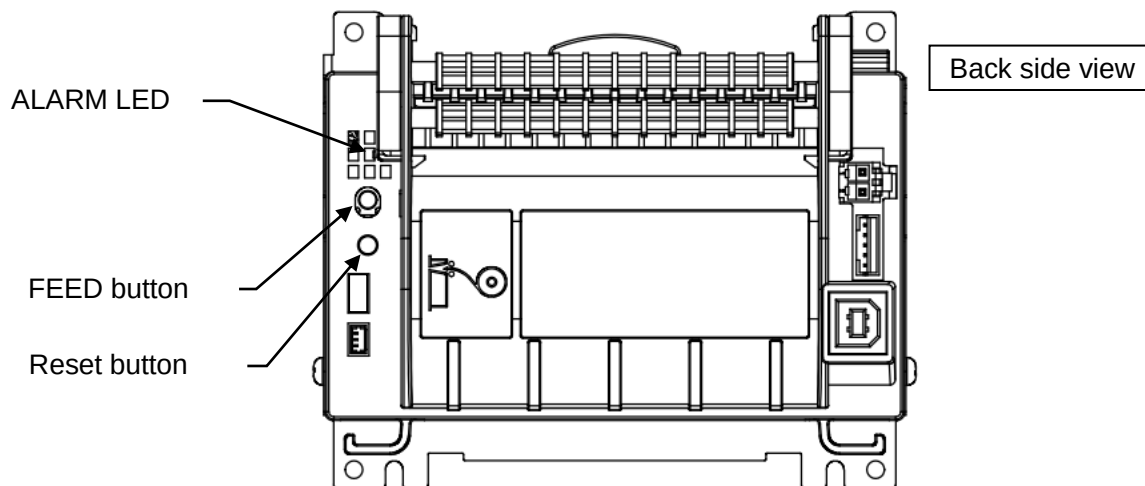
3) Reset button

Reset button is equipped under the FEED button.

(Please push it lightly with a tip of a ball-point pen, etc. and then release it.)

If reset is executed, settings must be done again since the printer will return to the state in which it was at the power on timing.

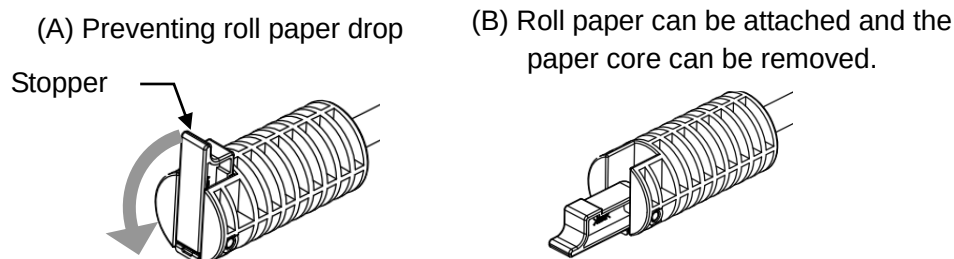
When the data remains in the buffer, please be careful of the data so as not to be deleted.



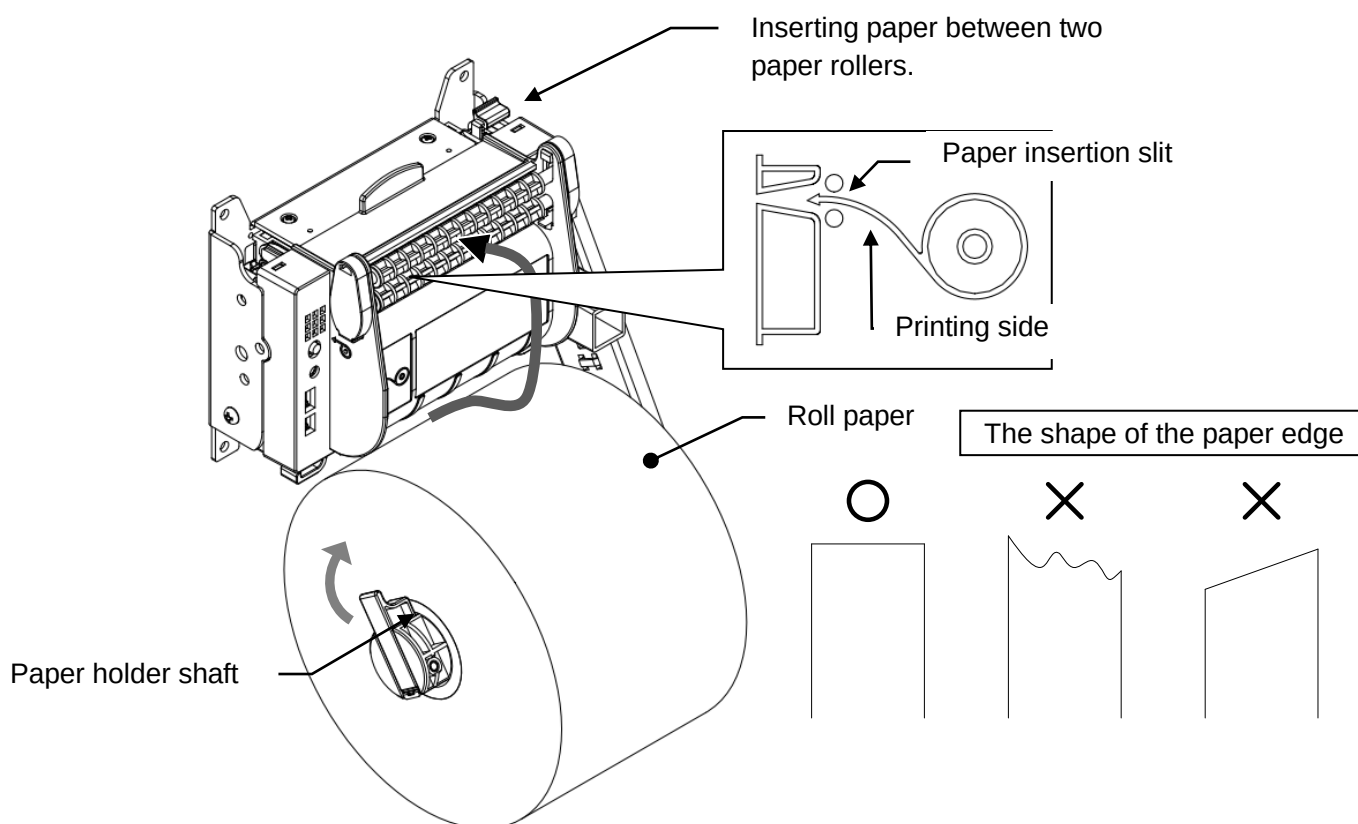
4.7 How to set roll paper

4.7.1 NP-F3092D (φ120_Arm)

- (1) Please turn ON the power.
- (2) Please move the stopper to the position indicated by the figure (B).



- (3) Please make the paper core penetrate through the paper holder shaft after checking the roll paper winding direction. (Please follow the figure below which is printed on the printer regarding the paper winding direction).
- (4) Please return the roll paper stopper position to the position indicated at in (A) above.
- (5) Please insert the tip of paper to the paper insertion slit (between two paper rollers) straightly.
- (6) Paper sensor will detect paper and paper will be pulled in automatically. (Please insert it securely until the retracting operation starts.)
- (7) Paper will be pulled in for fixed length, "TEST PRINT" (F/W Ver. 1.10 or later) will be printed, paper will be cut and the printer will become ready.

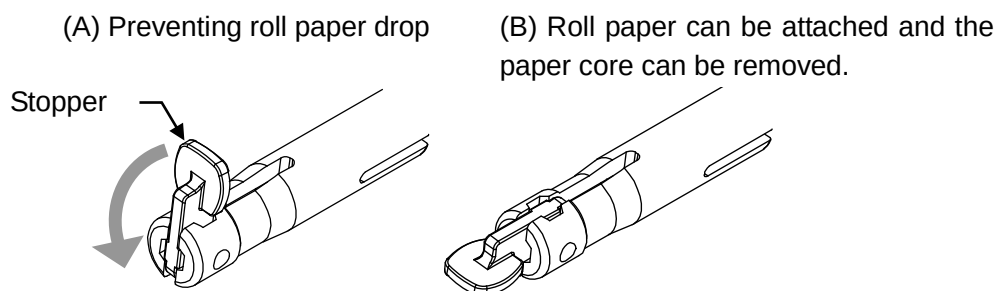


[Notes]

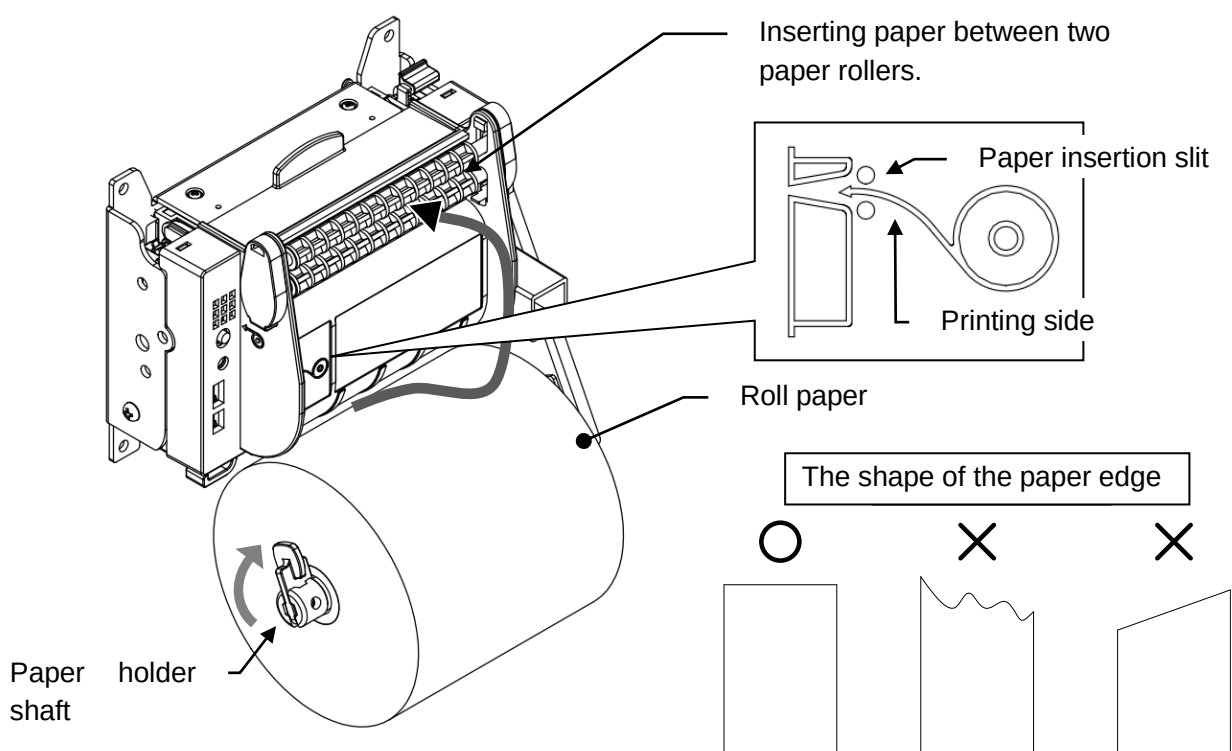
- Please DO NOT carry the printer with paper loaded.
- Please remove the old core before setting new roll paper.
- Please set paper correctly without loosening. (It causes paper jams.)
- Please set paper straight and at right angles. (Please refer to figure below for the shape of the paper edge.)
- When the edge of paper is fractured or cut diagonally, paper loading may not be operated.
- When the tip of paper is taped, please cut the taped part of the paper including the trace from which the tape was peeled off straight and with right angles by scissors, etc.
- When paper is inserted slantingly, please open the printer unit and straighten paper. (Please DO NOT apply excessive force beyond the stop position when opening the printer unit. Please beware not to get your hands or fingers pinched.)
- Since the thermal head may be highly-heated right after printing, please beware not to touch it by your hand or fingers, etc.

4.7.2 NP-F3094D (φ80_Arm)

- (1) Please turn ON the power.
- (2) Please move the stopper to the position indicated by the figure (B).



- (3) Please make the paper core penetrate through the paper holder shaft after checking the roll paper winding direction. (Please follow the figure below which is printed in the printer regarding the paper winding direction).
- (4) Please return the roll paper stopper position to the position indicated at (A) in the figure above.
- (5) Please insert the tip of paper to the paper insertion slit (between two paper rollers) straightly.
- (6) Paper sensor will detect paper and paper will be pulled in automatically.
(Please insert it securely until the retracting operation starts.)
- (7) Paper will be pulled in for fixed length, "TEST PRINT" (F/W Ver. 1.10 or later) will be printed, paper will be cut and the printer will become ready.



[Notes]

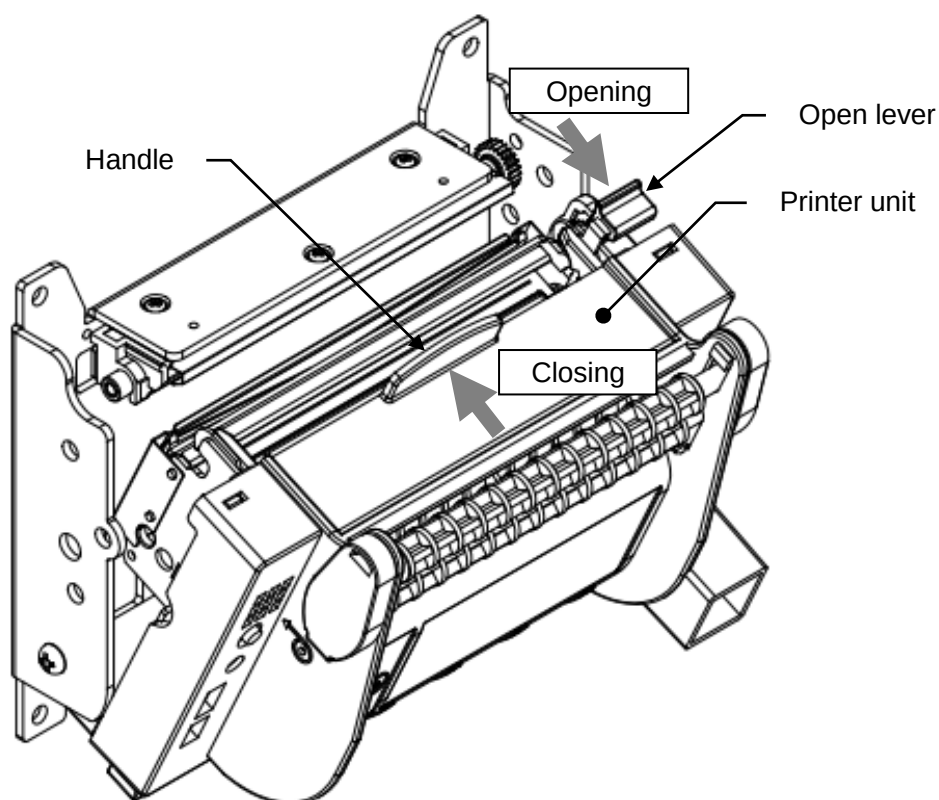
- Please DO NOT carry the printer with paper loaded.
- Please remove the old core before setting new roll paper.
- Please set paper correctly without loosening. (It causes paper jams.)
- Please set paper straight and at right angles. (Please refer to figure below for the shape of the paper edge.)
- When the edge of paper is fractured or cut diagonally, paper loading may not be operated.
- When the tip of paper is taped, please cut the taped part of the paper including the trace from which the tape was peeled off straight and with right angles by scissors, etc.
- When paper is inserted slantingly, please open the printer unit and straighten paper. (Please DO NOT apply excessive force beyond the stop position when opening the printer unit. Please beware not to get your hands or fingers pinched.)
- Since the thermal head may be highly-heated right after printing, please beware not to touch it by your hand or fingers, etc.

4.8 How to remove remaining paper and jammed paper

- 1) Please pull the open lever to the direction indicated by the arrow and open the printer unit by moving it.
- 2) Please remove all pieces of paper jammed on the paper path.
- 3) When closing the printer unit, please push the handle part and completely close the printer unit until it clicks and locks.

[Notes]

- Since the printer unit cannot be opened when the cutter blade is out of the standby position, please turn the power off and on to make it go back to the position.
- Since the thermal head may be highly-heated right after printing, please beware not to touch it by your hand or fingers, etc.
- Please DO NOT touch the metal edge part while the printer unit is opened.
- When closing the printer unit, please push the knob part at the center part and completely close it.
- Please be sure not to apply excessive force beyond the stop position when opening the printer unit.
- Please beware not to get your fingers or a hand pinched.



4.9 How to clean parts of the printer

In case wisps of paper, etc. adhere to the heat element of the thermal head, printing quality may deteriorate. Furthermore, paper dust may adhere to the platen roller and the sensor. In such cases, please turn OFF the power, open the printer unit and clean them as described below.

Please refer to “4.8 How to remove remaining paper and jammed paper” for opening and closing the printer unit.

1) Thermal head

Please clean the surface of the heating element with a cotton swab moistened with ethanol or isopropyl alcohol (IPA). (Please beware not to touch other parts.)

2) Platen roller

Please remove trash and dust, etc. on the surface by wiping the platen roller gently like rubbing with a dry cloth. (Please rotate the platen roller with your fingers in order to remove whole dust and dirt, etc.)

3) Paper sensor, paper near end sensor and around them

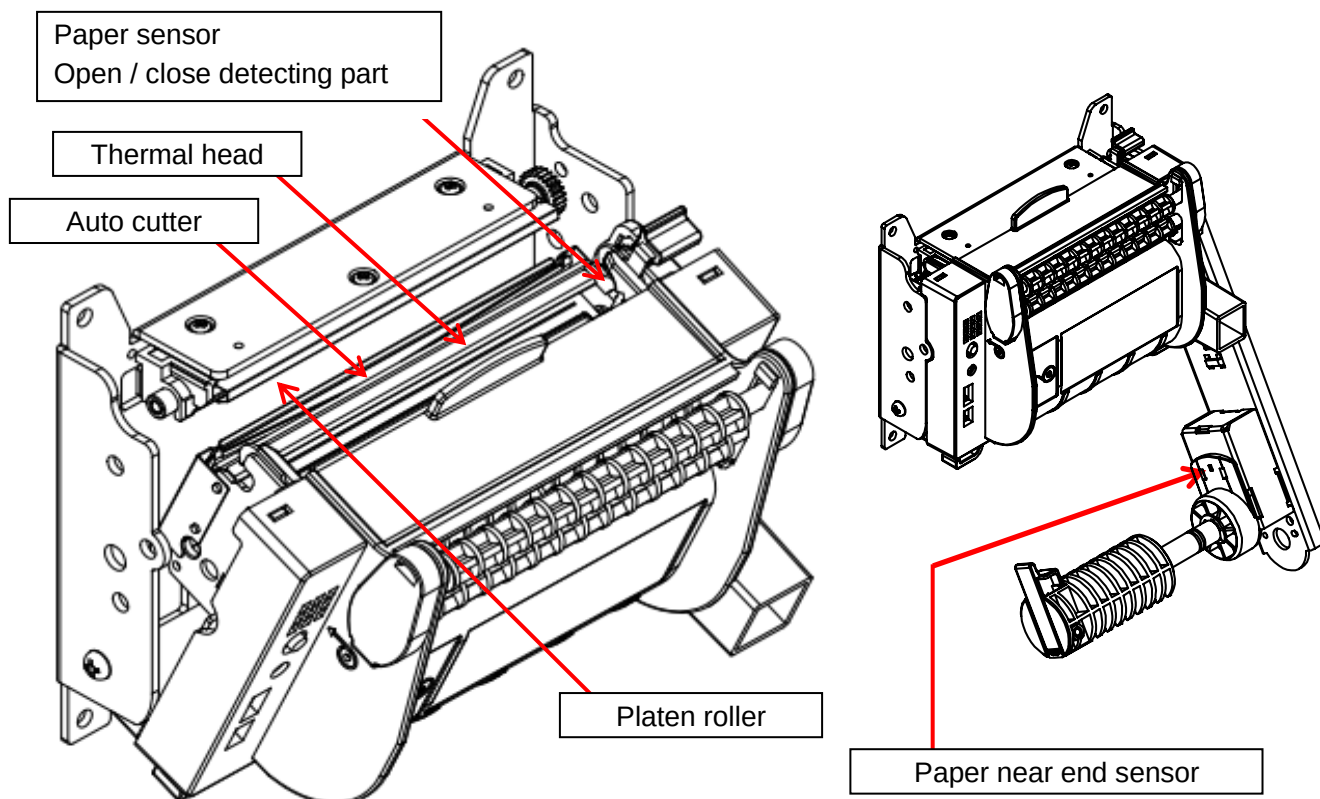
Please remove trash and dust, etc. adhered to the sensor with a soft bristle brush or a cotton swab.

4) Auto cutter

Please remove trash and dust, etc. adhered to the cutter with an air blow.
(Recommended frequency: every 100,000 operations)

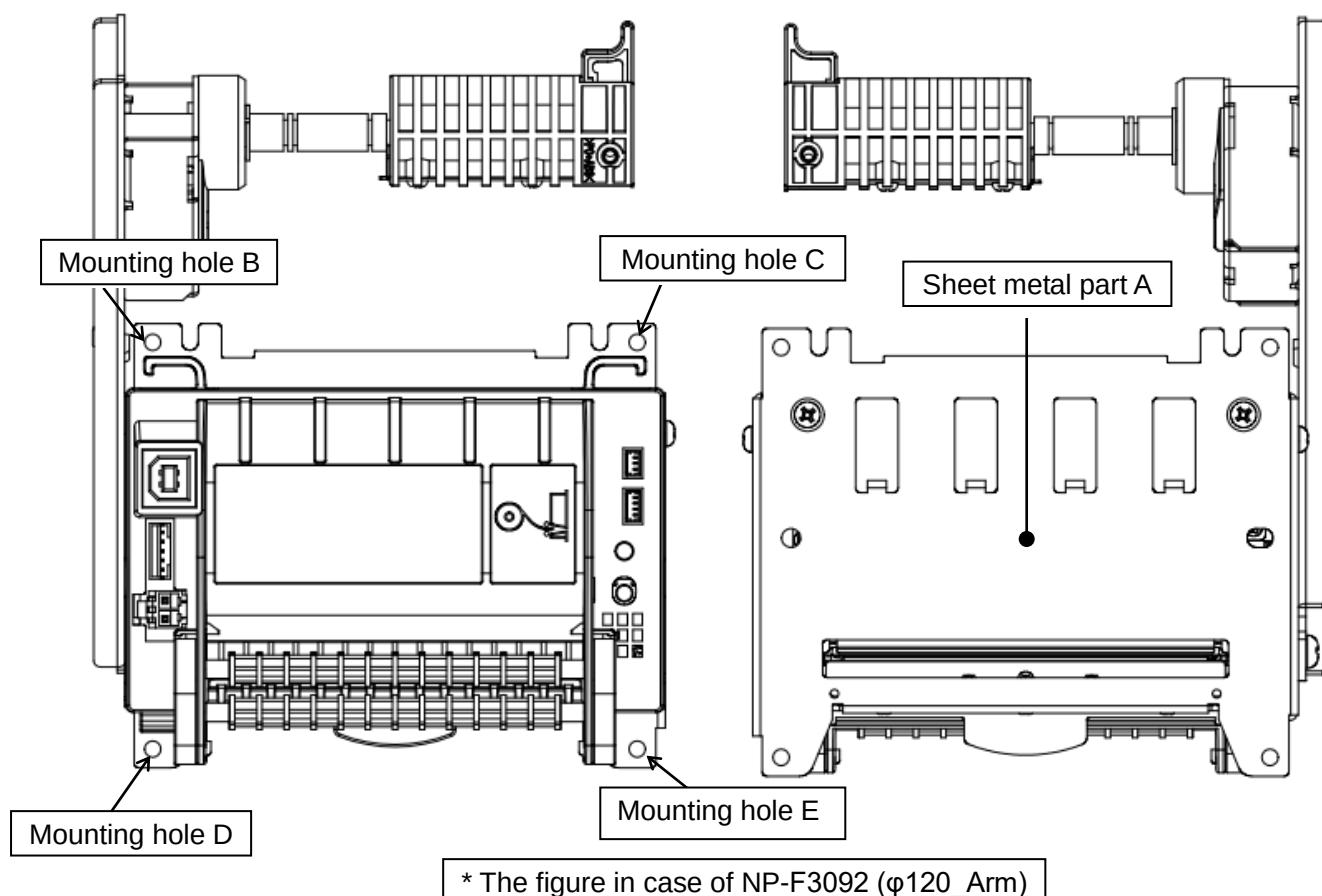
[Notes]

- Since the thermal head may be highly-heated right after printing, please beware not to touch it by your hand or fingers etc.
- Please DO NOT touch the heating element of the thermal head directly by your hand or with metal, etc.
- Please be careful of static electricity while cleaning since it may damage the thermal head.
- Since there is a case that paper dust easily appears depending on paper, please clean it periodically.
- Please be sure to turn ON the power after ethanol or isopropyl alcohol (IPA) dries completely.
- Please be sure NOT to apply excessive force beyond the stop position when opening the printer unit.



4.10 Installation

Please install this product after connecting the frame ground (FG) of this product and the frame ground (FG) of the housing in order to prevent malfunctions and the damage to the thermal head and the circuit board due to static electricity.



- 1) When the mounting part of the housing is a frame ground (sheet metal), please connect the frame ground by contacting the frame ground (sheet metal part A) of this product with the frame ground of the mounting part of the housing.
- 2) When the mounting part of the housing is not a frame ground (sheet metal), please connect this product and the frame ground of the housing by using the cable with a terminal (AVG#20 or thicker) and a screw (A screw with a washer is recommended.) to mounting hole B, C, D or E.

[Notes]

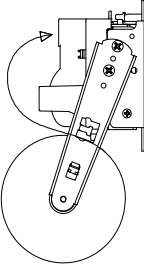
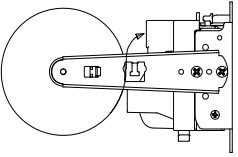
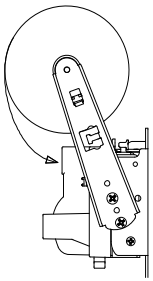
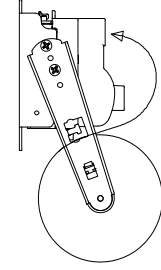
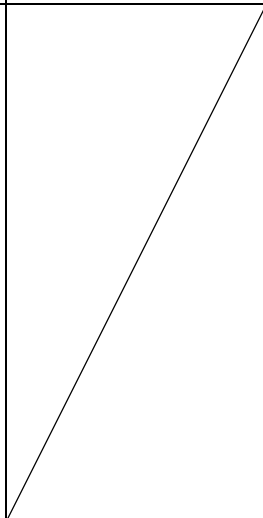
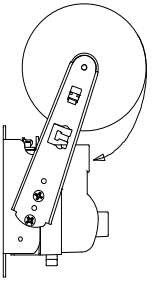
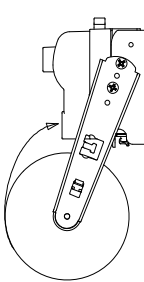
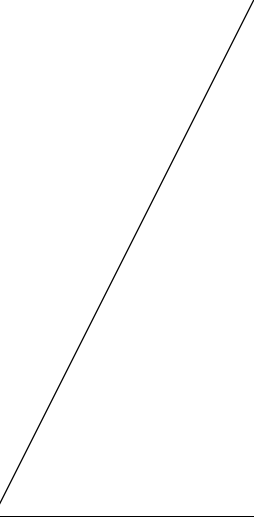
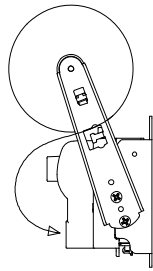
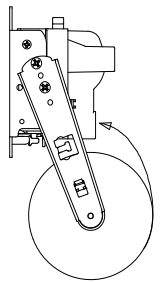
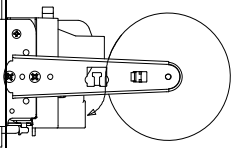
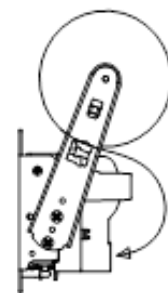
- Using a washer for connecting the terminal to mounting hole B, C, D or E will make the connection to the frame ground tighter. (Please check whether there is conduction to the housing.)
- Please install this product without applying excessive force, force generating deformation or twisting movement.
- If force generating deformation or twisting movement is applied, it may be the cause of printing quality deterioration, serpentine thermal paper feeding, paper jam, noise generation and improper cutting.
- Please set this product at the flat surface where vibration will not easily occur.
- Please be careful not to damage the lead wire, etc. when fastening screws.
- Please take the space for connecting cables to connectors since there are connectors for power supply and communication.
- Cut paper will not fall sometimes even when this printer is used with full cut mode. Please design the exterior so that paper can be taken out regardless of the paper condition.
- Please design this product's installation so that it will not receive any interferences from exterior components. If it receives interferences, it will affect printing or cutting and it will be the cause of the damage to the printer.

5. Paper holder

5.1 Selecting the roll paper position and whether printing side will be ejected upward or downward

* The figure in case of NP-F3092 (φ120_Arm)

This printer can be installed by 10 ways described below. Please refer to the table below and select the roll paper position and whether printing side will be ejected upward or downward after considering which is suitable for the terminal device.

		In case that roll paper will be set from the left side viewed from the back, the paper holder unit should be set to the right side.			In case that roll paper will be set from the right side viewed from the back, the paper holder unit should be set to the left side.		
		Roll paper "lower"	Roll paper "middle"	Roll paper "upper"	Roll paper "lower"	Roll paper "middle"	Roll paper "upper"
Ejecting paper with the printing surface downward (recommend)	Factory default setting						
Ejecting paper with the printing surface upward							

[Notes]

- Ejecting paper with the printing side downward is recommended since paper setting and removing jammed paper, etc. are more operable and easier.
- Please DO NOT use this printer with its position slanting.
- Although paper may bend downward by using paper with the printing surface upward and with roll paper "lower", there will be no problem with printing operation.

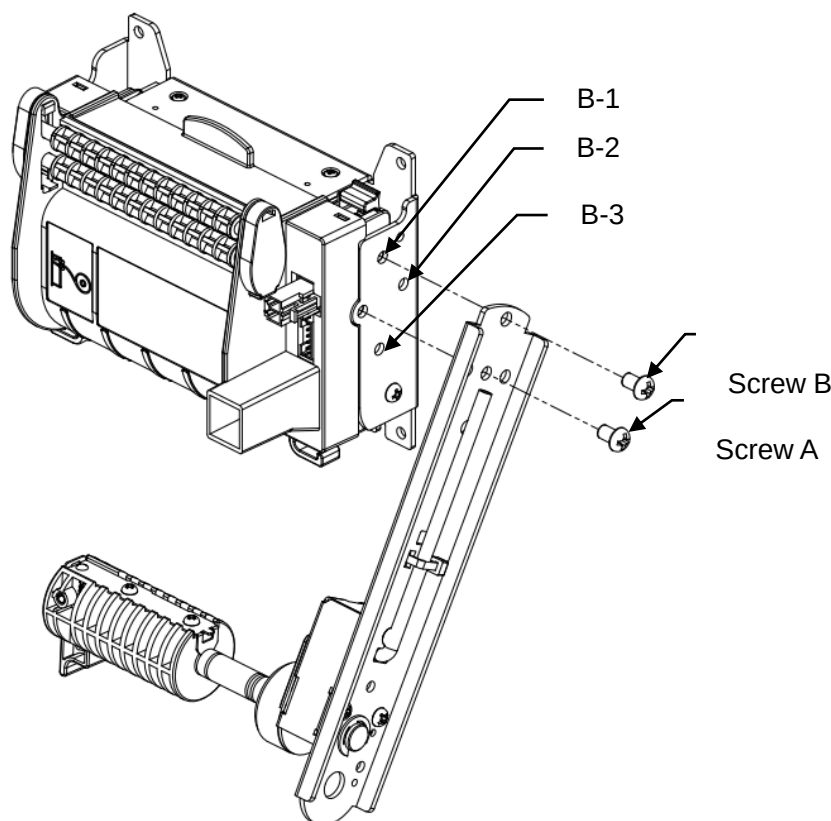
5.2 Changing the paper holder position

1) When changing the position to another on the same side.

- Please unscrew the screw B (M4).
- Please loosen the screw A (M4) at the fixed position (rotation center) slightly.
- Please rotate the paper holder with the screw A at the center to the position with screw B at B-1 mounting hole, screw B at B-2 mounting hole or screw B at B-3 mounting hole and fix it with the screw B.
- Please tighten the screw A.

2) When changing the position to one on the other side.

- Please disconnect the paper near end sensor cable.
- Please detach the paper holder by unscrewing screw A (M4) and screw B (M4).
- Please attach the paper holder temporarily on the other side by using the screw A (M4) with temporary tightening.
- Please rotate the paper holder with the screw A at the center and fix it with the screw B to the mounting hole B-1 or B-3.
- Please tighten the screw A.



* The figure in case of NP-F3092 ($\phi 120_Arm$)

[Notes]

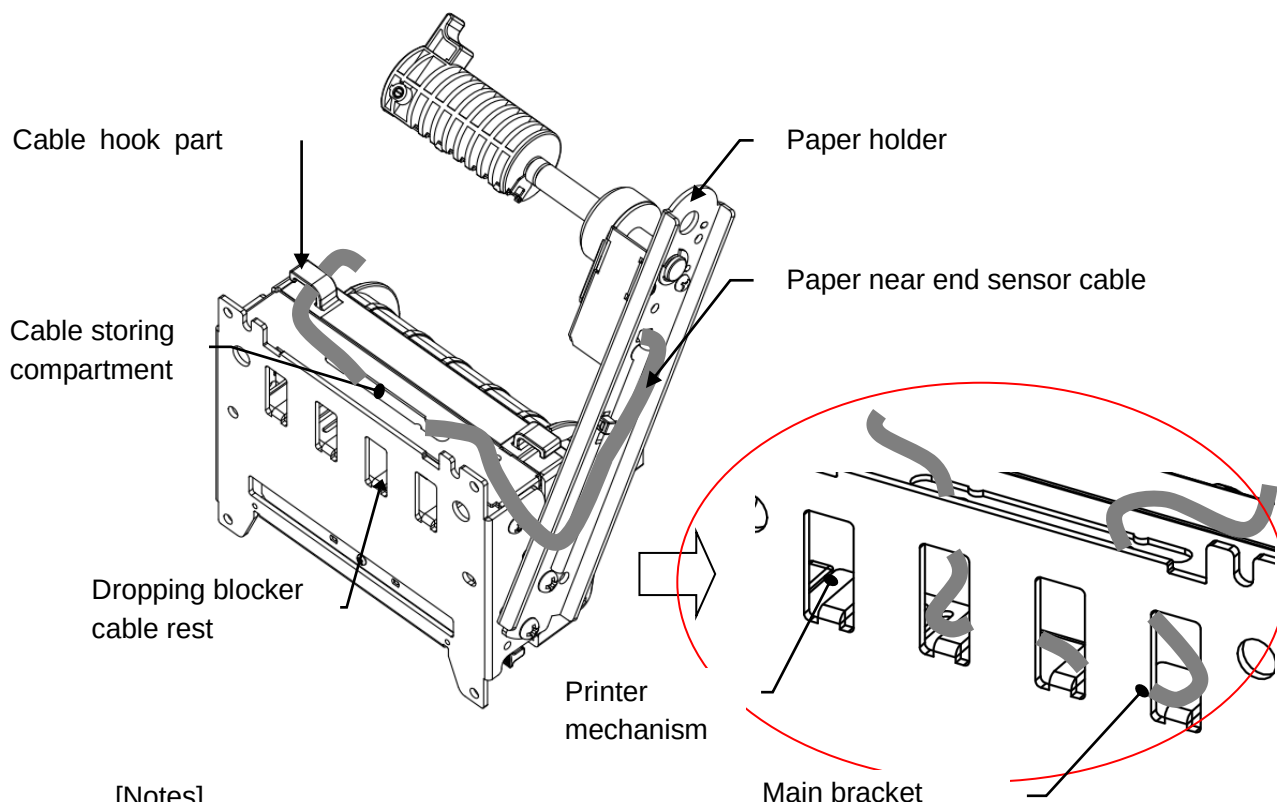
- Please arrange the paper near end sensor cable location appropriately.
- Screws (M4) are self-tapping type. When screwing to the new attachment hole, the force for screwing it may be strong.
- Although attaching positions of the paper holder are symmetric, there is no B-2 hole on the left side (only 2 holes of B-1 and B-3).
- Please use a Phillips screwdriver (#2).
- Please DO NOT use the screws other than the attached screws (A, B and C). If the length or the kind of screws is different from the attached screws, it may be causes of damage. If the screws are lost, please contact us.

5.3 How to handle cables

* The figure in case of NP-F3092 (φ120_Arm)

1) How to handle the paper near end sensor cable

Depending on the paper holder attaching position, the cable length may be left over. There is a cable storing compartment equipped to the position described in the figure below. In case the cable length is left over, please use it when needed.



[Notes]

- Please be sure to lay the cable on the dropping blocker cable rests inside the cable storing compartment. Please **DO NOT** push the cable into the space between the printer mechanism and the main bracket so as not to make printer unit opening and closing disturbed.

2) How to handle the interface cable and the power cable

Depending on the paper holder attaching direction and the roll paper position, how to handle cables varies. Please wire and store them after taking maintenance works into consideration so that they will not hindrances when replacing roll paper, etc.

There is a cable hook part equipped at the position described in the figure below. Please use it when needed.

